

Structural Unification of Gravity and Quantum Dynamics under AASC

Einstein and Schrödinger Dynamics as Forced Paired Projections of the Admissible Physical Interior

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Abstract

This manuscript proves a structural unification theorem under AASC. It does not quantize gravity, geometrize the wavefunction, or treat either downstream formalism as primitive. Gravity is derived first as the admissibility-forced constraint role of the physical interior. Quantum dynamics is derived as the projective-Hilbert and record-bearing continuation projection of that same admissible interior. Einstein dynamics and Schrödinger dynamics are therefore not primitive rivals, and neither one is a foundation for the other. By singleton admissible-interior closure, once the physical-interior target is fixed there is one admissible interior up to lawful equivalence. The paired-projection claim is therefore not a coordination of two admissible interiors; the metric and projective-Hilbert projections are faithful projections of that one upstream AASC structure.

The proof begins from a determinate nondegenerate physical-interior target. Such a target forces the AASC kernel

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

with admissibility separating licensed use from hidden import, standing supplying reusable theorem-bearing force, reference fixing same-domain target identity, and irreversibility blocking same-act repair. The fixed-domain consequence layer blocks primitive metric backflow, primitive Hilbert backflow, hidden representative privilege, source-side laundering, record-side laundering, compatibility laundering, branch-selection import, and undeclared carrier transfer. From this common target, the gravity side is forced as structural constraint architecture and the metric projection yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The quantum side is forced as projective-Hilbert and record-bearing continuation, and the scalar nonrelativistic first-response projection yields

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

These equations are paired theorem objects of the same admissible physical interior under different faithful projections. The quantum normal form used here is the closed scalar nonrelativistic projective-Hilbert and record-bearing first-response theorem; spin, gauge, relativistic, field-theoretic, open-system, and measurement/update structures are typed as declared richer, effective, or update scopes.

The joint-ledger theorem supplies the cross-projection adequacy condition, not the source of unity. The structural unity is upstream; the hard problem is downstream cross-projection closure. Whenever metric source standing and quantum transition/record standing interact, admissible coupling must preserve source closure, record fixation, boundary compatibility, lawful redescription, joint continuation, and reportability through a declared joint ledger. Measurement

and record formation are controlled by the admissible-record construction theorem. Entanglement and nonlocality are controlled by the Bell-scope compatibility theorem: entangled preparations carry joint tensor compatibility, not hidden metric transport, superluminal communication, source transfer, or same-scope repair.

The central cross-projection result is the same-scope joint-coupling exhaustion theorem. Every attempted GR/QM cross-coupling is either a declared joint ledger or one of the classified failure/scope-change routes: projection backflow, source laundering, record laundering, compatibility laundering, measurement/update scope, open/effective scope, boundary/certificate route, richer-scope joint continuation, or standing failure. Semiclassical gravity, QFT on curved spacetime, collapse/stochastic gravity, horizon information claims, holographic reconstruction, canonical quantum gravity, and path-integral routes are therefore typed by the ledger they actually carry rather than dismissed or overpromoted.

The reportable endpoint is structural unification rather than an exhaustive computation of all richer-scope dynamics. Gravity is not a quantum target. Gravity is the structural constraint role of admissible physical construction. Quantum dynamics is a faithful projection, not the ground of gravity. Quantum gravity, properly typed, is the problem of richer-scope joint continuation preserving both projections, not primitive quantization of the gravitational role.

Keywords: AASC; structural unification; singleton admissible interior; no plural interiors; gravity as structural necessity; quantum dynamics; Einstein equation; Schrödinger equation; quantum gravity; admissibility; standing; projection; joint ledger; source closure; transition probability; record fixation; entanglement; boundary compatibility; semiclassical gravity; QFT on curved spacetime; measurement; horizon information.

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Read This First: Structural Unification Claim and Proof Spine

The proof class is *kernel-forced structural unification by paired projection*. Its correctness is determined by the declared target, hypotheses, named proof links, and reportable scope. It is not evaluated by resemblance to metric-first quantum gravity, canonical quantization, path-integral quantization, semiclassical gravity, information-theoretic reconstruction, or Hilbert-space-first foundational programs.

The exact structural claim is:

Gravity and quantum dynamics are structurally unified because both are forced projections of one AASC standing-preserving physical interior.

The anti-split claim is:

There is not one admissible interior for gravity and another for quantum standing.
There is one admissible interior with multiple faithful downstream projections.

The cross-projection claim is:

When the metric and projective-Hilbert projections interact,
a declared joint ledger is required to preserve source closure, transition standing,
record fixation, boundary compatibility, and reportability without backflow.

The proof spine has three layers.

Structural unification layer.

determinate nondegenerate physical-interior target
 $\Rightarrow K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}) \Rightarrow A^+$
 $\Rightarrow$ standing-instantiated admissible regime
 $\Rightarrow$ unique admissible physical interior
 $\Rightarrow \begin{cases} \text{gravity as structural constraint necessity,} \\ \text{quantum standing as projective/record-bearing continuation.} \end{cases}$

Paired projection layer.

$$\text{AASC gravity role} \Rightarrow \Pi_g \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}},$$

$$\text{AASC quantum standing role} \Rightarrow \Pi_H \Rightarrow i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

Joint-ledger interaction layer.

$$(\Pi_g, \Pi_H) + \text{cross-projection use} \Rightarrow \mathcal{L}_{gH},$$

$$\mathcal{L}_{gH} = (\mathcal{L}_g, \mathcal{L}_H, \mathcal{L}_{\text{cross}}, \mathcal{L}_{\text{rec}}, \mathcal{L}_{\text{comp}}, \mathcal{C}_{\text{cert}}, \text{Cont}_{\text{joint}}, \text{Rep}).$$

No declared joint ledger means that a purported coupling is projection backflow, source laundering, record laundering, compatibility laundering, open/effective scope, measurement/update scope, boundary/certificate route, richer scope, or standing failure.

This manuscript does not retreat from unification. It distinguishes structural unification from completed richer-scope dynamics. The unity is upstream in the admissible physical interior; the hard work is downstream cross-ledger closure.

This paper's three normal-form predicates align as follows:

Einstein dynamics	$\text{SameScopeMetric}_1(E_{\mu\nu})$	$\Rightarrow$ Einstein metric/source form,
Schrödinger dynamics	$\text{SameScopeQuantumSpatial}_1(H)$	$\Rightarrow H = a\Delta + M_V,$
Structural unification	$\text{SameScopeJoint}_1(C)$	$\Rightarrow C \in \mathcal{L}_{gH}.$

Roadmap and Reportable Scope

The manuscript proceeds in three movements. First, it fixes the kernel-forced physical-interior target and shows why gravity is structural necessity rather than a quantum target. Second, it uses the Einstein and Schrödinger normal-form results as paired theorem-use ladders: one metric, one projective-Hilbert/record-bearing. Third, it proves the cross-projection ledger theorem: once both projections are used in one target, interaction must be carried by a declared ledger preserving metric source closure, quantum transition standing, record fixation, boundary compatibility, and reportability together.

The reportable result is:

structural unification at the AASC physical-interior level + cross-projection ledger closure.

The manuscript does not report all richer-scope quantum-gravity dynamics, numerical constants, all matter sectors, all measurement dynamics, all QFT, or all black-hole endpoint physics. It does report why gravity is not a primitive quantization target and why Einstein and Schrödinger dynamics are paired forced projections rather than primitive rivals.

Part I

Standalone AASC Foundation and Structural Unification Target

1 Proof-class lock

The proof class is a paired-projection theorem with a joint-ledger endpoint. A criticism is correctness-relevant only when it attacks a named proof link: target instantiation, kernel necessity, quotient architecture, metric projection, projective-Hilbert projection, Einstein normal form, Schrödinger normal form, projection non-primacy, joint-ledger adequacy, no-backflow discipline, or reportable-scope discipline.

Definition 1.1 (Correctness-relevant criticism). *A criticism is correctness-relevant to this manuscript iff it identifies a false premise, invalid inference, hidden import, scope shift, over-reported conclusion, or broken named link in one of the following locations:*

1. determinate physical-interior target adequacy;

2. *kernel forcing from target adequacy;*
3. *fixed-domain consequence layer A^+ ;*
4. *quotient architecture and standing-preserving continuation;*
5. *faithful metric projection theorem-use;*
6. *faithful projective-Hilbert projection theorem-use;*
7. *projection non-primacy and no-backflow discipline;*
8. *joint-ledger definition and adequacy predicate;*
9. *classification of existing GR/QM interface routes;*
10. *paired normal-form theorem;*
11. *joint-ledger necessity theorem;*
12. *reportable-scope statement.*

Remark 1.2 (Proof-class mismatch). *A demand that this argument first become canonical quantum gravity, covariant path-integral quantum gravity, semiclassical Einstein theory, string theory, loop quantum gravity, emergent spacetime, objective collapse, or many-worlds interpretation changes the proof class unless it identifies a broken link in the paired-projection theorem. Those programs are classified later as interface routes. They are not the correctness standards for this theorem unless they provide a same-target counterexample.*

Proposition 1.3 (Proof-class boundary). *Within the declared proof class, the relevant correctness question is whether the same physical interior forces the kernel, admits two faithful first-response projections, blocks projection backflow, and requires a declared joint ledger for GR/QM coupling. Preference for a different quantum-gravity program has no bearing unless it locates a failure in those links.*

2 Kernel-neutral target adequacy and the AASC kernel

The physical-interior target is not a verbal label. It is a theorem-bearing object whose metric and quantum projections are supposed to be about the same admissible physical content. Such an object must have target identity, standing status, admissible continuation, lawful redescription, and failure behavior before either the metric or Hilbert presentation can carry theorem force.

Definition 2.1 (Kernel-neutral target adequacy). *A theorem-bearing act is target-adequate when it satisfies:*

1. **Target determinacy:** *the act has a repeatable object of evaluation, not a presentation-dependent trace.*
2. **Step/status evaluability:** *assertions and transitions are evaluable as advancing, blocking, or failing relative to the same target.*
3. **Act-time finality:** *later repair or reclassification is a distinct act, not retrospective standing for the same act.*

4. **Same-regime fidelity:** lawful redescription preserves target, carrier, role, standing status, and continuation profile.

Definition 2.2 (AASC kernel). *The AASC kernel is*

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

where *Adm* distinguishes licensed from unlicensed use, *Stand* supplies reusable theorem-bearing force, *Ref* fixes same-domain target identity, and *Irr* blocks same-act post-hoc repair.

Theorem 2.3 (Target adequacy forces the kernel). *If a theorem-bearing physical-interior act is target-adequate, then $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$ is already in force for that act.*

Proof. Target determinacy requires a role that preserves the same object of evaluation across use, comparison, and lawful redescription; that role is reference. Step/status evaluability requires a role by which a statement or transition can carry licensed force for later use; that role is standing. Act-time finality requires a role preventing later intervention from changing the status of the same act; that role is irreversibility. Same-regime fidelity requires a boundary under which these classifications remain jointly enforceable; that boundary is admissibility. Thus the kernel is not added by vocabulary. It is the minimal governance work already performed by any target-adequate theorem act. $\square$

Corollary 2.4 (Vocabulary independence). *A critic may reject the names $\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}$, but if the critic preserves determinate target identity, theorem-bearing status, same-act failure, lawful redescription, and continuation profile, the critic preserves the kernel roles under other names. If any role is denied, the same theorem-bearing physical-interior target is lost.*

Proposition 2.5 (Anti-weakening burden). *A proposed replacement for the kernel is correctness-relevant only if it preserves the same target, theorem-bearing force, act-time finality, lawful redescription, continuation profile, and reportability while omitting one of the kernel roles and without performing the same governance work under another name.*

3 Fixed-domain consequence layer A^+

Write

$$\text{nondegenerate determinate same-domain use} \Rightarrow K \Rightarrow A^+.$$

For the joint GR/QM theorem, A^+ contains:

1. **AMetric no-selector boundary:** no metric, Hilbert norm, probability, wavefunction, Hamiltonian, curvature, scale, basis, coordinate, boundary datum, or source number has primitive admissibility authority.
2. **No second same-scope classifier:** a factor that changes theorem standing is anchor motion, declared tensor/operator load, or forbidden hidden selector work; if it changes no standing, it is skin.
3. **No same-act repair:** later collapse, normalization, renormalization, source completion, horizon rescue, branch selection, or record insertion is a new act or scope change.
4. **Standing as reuse-stable admissibility:** standing is admissibility preserved under licensed same-domain continuation.

5. **No representative privilege:** coordinates, frames, bases, global phase, chart labels, branch labels, and presentation choices carry no physical standing unless declared as anchors or tensor/operator content.
6. **No projection backflow:** a downstream metric fact cannot authorize quantum standing, and a downstream Hilbert fact cannot authorize metric standing.
7. **Report preservation:** a report may not claim more theorem force than the act carries.
8. **No carrier transfer without declaration:** moving between metric, Hilbert, field-theoretic, measurement, boundary, or semiclassical carriers changes scope unless declared.

Theorem 3.1 (No independent same-scope joint classifier). *In a fixed nondegenerate physical-interior theorem domain, no independent same-scope classifier can decide GR/QM standing outside the admitted interface. Any proposed classifier is governance-equivalent to the admitted interface, skin, declared anchor motion, declared tensor/operator load, or forbidden hidden selector work.*

Proof. If the classifier changes standing-bearing use, it performs admissibility-relevant work. In a fixed domain such work is already governed by the kernel roles. If it preserves those roles exactly, it is equivalent. If it changes carrier, source, boundary, connection, Hilbert structure, record structure, or continuation rule, it moves anchors or adds tensor/operator load. If it changes no standing-bearing use, it is skin. The remaining case is undeclared same-scope authority, which violates admissibility. $\square$

4 Cost of kernel denial in the joint theorem

Kernel denial does not preserve the same GR/QM structural-unification theorem with fewer assumptions. It changes or destroys the determinate paired-projection object.

Kernel role denied	Joint GR/QM cost
Reference	No fixed physical-interior target; metric and quantum carriers cannot be compared as projections of the same object. The symbols $g_{\mu\nu}$, $T_{\mu\nu}$, ψ , H , records, sources, and boundary data no longer have same-domain identity.
Standing	Einstein and Schrödinger equations become parallel syntax or modeling conventions, not theorem-bearing continuations of a common physical interior.
Admissibility	No boundary separates metric source ledger, quantum transition ledger, measurement update, semiclassical averaging, QFT enrichment, branch selection, or hidden selector import.
Irreversibility	Collapse, renormalization, source repair, decoherence, horizon recovery, record creation, or branch labeling can be laundered into same-act standing.

Corollary 4.1 (No third position). *There is no position from which one may reject the kernel while still treating Einstein and Schrödinger dynamics as determinate paired projections of one physical-interior theorem object. Either the paired target remains nondegenerate, in which case the kernel is active, or one of the costs in the table is paid and the same theorem object is lost.*

5 Singleton interior closure and the no-split burden

The structural-unification theorem does not posit a gravity-side admissible interior and a quantum-side admissible interior and then coordinate them externally. Once the physical-interior target is fixed by admissibility boundary, standing classification, target fixation, and admissible continuation, the admissible interior is unique up to lawful equivalence. A purported second admissible interior is not a same-domain alternative. It is anchor motion, scope change, competing gate import, carrier laundering, or skin-level duplication. This is the theorem-use role of the singleton admissible-interior closure result [12].

Theorem 5.1 (No plural-interior split). *Let $\mathcal{I}$ be the fixed admissible physical interior for the structural-unification target. Then there do not exist two same-domain admissible interiors*

$$\mathcal{I}_g \quad \text{and} \quad \mathcal{I}_H$$

such that gravity derives from $\mathcal{I}_g$, quantum standing derives from $\mathcal{I}_H$, and the two are merely externally coordinated on the same fixed domain. Any purported split falls into at least one of the following typed routes:

1. $\mathcal{I}_g = \mathcal{I}_H$ up to lawful equivalence;
2. one of the two is skin-level duplication;
3. the split imports a competing gate or standing classifier;
4. the split changes domain or scope;
5. the split imports a carrier, generator, or repair route;
6. standing fails.

Therefore $\mathcal{I}_g \neq \mathcal{I}_H$ is not a same-domain admissible alternative. The admissible same-domain form is

$$\mathcal{I} \Rightarrow (\Pi_g(\mathcal{I}), \Pi_H(\mathcal{I})).$$

Proof. A same-domain split into $\mathcal{I}_g$ and $\mathcal{I}_H$ would require either two admissible interiors under one fixed admissibility boundary and one standing classification, or two competing standing classifiers. The singleton closure theorem eliminates the first route: once the standing-instantiated admissible regime is fixed, discrete plural admissible interiors collapse extensionally or fail. The no-competing-gates result eliminates the second route: two classifiers that disagree on the same fixed domain introduce indeterminate standing or a boundary selector. If the proposed split changes the admissibility envelope, target fixation, continuation rule, carrier, or boundary condition, it changes scope rather than preserving the same interior. If it changes none of these, it is skin-level duplication. Carrier, generator, and repair routes are excluded by the same singleton-closure package. The listed cases exhaust the same-domain alternatives, so no plural-interior split remains. $\square$

Corollary 5.2 (One interior, paired projections). *If Π_g and Π_H are faithful projections in the structural-unification target, then they are projections of one admissible physical interior:*

$$\Pi_g : \mathcal{I} \rightarrow \mathcal{G}, \quad \Pi_H : \mathcal{I} \rightarrow \mathcal{Q}.$$

They are not maps from two admissible interiors unless those interiors coincide up to lawful equivalence.

Proof. If the projections came from two same-domain admissible interiors, the no plural-interior split theorem would force those interiors to be lawfully equivalent. Otherwise the construction is a scope split, hidden classifier import, carrier laundering, or standing failure. Hence the paired-projection form is one interior with two faithful downstream projections. $\square$

Part II

The AASC Physical Interior and Structural Unification Target

6 Admissible physical interior

Definition 6.1 (Admissible physical interior). *An admissible physical interior is a determinate same-domain regime supporting standing-bearing physical contents whose lawful continuation preserves target identity, whose redescriptions preserve quotient identity, whose failures cannot be repaired as the same act, and whose reusable roles are typed as anchors, tensors/operators/compatibilities, or skin.*

Definition 6.2 (Singleton paired AASC physical interior). *A singleton paired AASC physical interior is*

$$\mathcal{I}_{gH} = (\mathcal{I}, \text{Anc}, \text{Ten}, \text{Skin}, \sim, \text{Cont}, \Pi_g, \Pi_H, \mathcal{L}_{gH}),$$

where $\mathcal{I}$ is the unique admissible interior fixed by the target, Anc fixes the admissibility envelope, Ten carries standing-bearing invariant content, Skin is presentation surplus, $\sim$ is lawful redescription, Cont is admissible continuation, Π_g is the faithful metric projection, Π_H is the faithful projective-Hilbert projection, and $\mathcal{L}_{gH}$ is present when both projections are jointly used. No second same-domain admissible interior is available as a repair or coordination device.

Definition 6.3 (Same-interior fidelity). *The paired projections have same-interior fidelity iff*

$$x \sim_{\mathcal{I}} y \Rightarrow \Pi_g(x) \sim_g \Pi_g(y)$$

and

$$x \sim_{\mathcal{I}} y \Rightarrow \Pi_H(x) \sim_H \Pi_H(y),$$

and admitted continuation in $\mathcal{I}_{gH}$ projects to standing-preserving metric, quantum, and declared joint continuation without projection backflow.

Theorem 6.4 (Same-interior fidelity). *If Π_g and Π_H are faithful paired projections of the structural-unification target, then they preserve lawful redescription and continuation from the same singleton interior. If either projection fails to preserve $x \sim_{\mathcal{I}} y$ in its target presentation, then it is not a faithful projection of the same interior. The failure is anchor motion, hidden selector import, competing gate import, scope split, or standing failure. Because the interior is unique, failed same-interior fidelity cannot be repaired by moving one projection to a second same-domain admissible interior.*

Proof. Faithful projection means that standing-bearing distinctions in the target presentation are images of standing-bearing distinctions in the source interior, while projection kernels are skin relative to the declared comparison class. If $x \sim_{\mathcal{I}} y$ but $\Pi_g(x) \not\sim_g \Pi_g(y)$, the metric projection has introduced a standing-bearing distinction not present in the source. If $x \sim_{\mathcal{I}} y$ but $\Pi_H(x) \not\sim_H \Pi_H(y)$,

the projective-Hilbert projection has done the same. Such an introduced distinction is a hidden selector, anchor motion, or scope change unless it is declared richer-scope tensor/operator load. The no-plural-interior theorem blocks the attempted repair in which the divergent projection is assigned to a second admissible interior. Thus same-interior fidelity is forced by faithful paired projection from the singleton interior. $\square$

Theorem 6.5 (Target-instantiated structural unity). *Once the target is a determinate nondegenerate physical interior bearing both metric deformation-response standing and quantum transition/record standing, paired projection capacity is not an optional interpretive overlay. It is part of the target instantiation: the same upstream admissibility, standing, reference, irreversibility, quotient identity, continuation discipline, and singleton admissible interior feed both faithful projections.*

Proof. The target is one physical-interior theorem object, not two unrelated notations. Reference fixes the same target; standing supplies reusable force; admissibility separates licensed projection from hidden import; irreversibility blocks later repair. Singleton admissible-interior closure blocks a same-domain split into separate gravity and quantum interiors. A metric projection and a projective-Hilbert projection may present different theorem objects, but if they concern the same physical interior, their redescription and continuation behavior must factor through the same upstream target. Otherwise the claim is not structural unification but juxtaposition of two models or a scope split. $\square$

7 Gravity is structural necessity, not a quantum target

Conventional quantum-gravity rhetoric often begins with

$$\text{GR is classical} \Rightarrow \text{quantize gravity.}$$

AASC reverses the order:

$$\text{admissible physical interior} \Rightarrow \text{gravity as constraint necessity} \Rightarrow \text{metric projection.}$$

Gravity is not waiting to be made quantum. Gravity is the structural constraint role required for admissible physical construction. Quantization is a different projection regime. Quantizing a metric projection cannot generate a second admissible interior for gravity; both the metric and projective-Hilbert projections are downstream of the singleton admissible interior.

Theorem 7.1 (No quantization-primacy theorem). *Let $\mathcal{I}$ be a determinate nondegenerate AASC physical interior. The gravitational role of $\mathcal{I}$ is fixed as constraint architecture before metric field-equation form and before projective-Hilbert quantum projection. Therefore no projective-Hilbert operation, quantization procedure, path integral, canonical operator assignment, or Hilbert-space representation can authorize gravity as a same-scope primitive. If quantum formalism is used to authorize gravity itself, the move is projection backflow.*

Proof. The kernel and A^+ force admissibility, standing, reference, irreversibility, quotient architecture, lawful redescription, and continuation before either the metric or projective-Hilbert projection can carry theorem force. The AASC gravity theorem fixes gravity as the constraint role of the physical interior. Metric dynamics is a faithful projection of that role. Quantum dynamics is a different faithful projection. A downstream projection cannot authorize the upstream structure that licensed it. Thus treating quantization as the primitive authorization of gravity reverses the dependency order and imports projection backflow. $\square$

Corollary 7.2 (Category error of primitive quantization). *The category error is not failure to quantize gravity. The category error is treating gravity as a downstream object awaiting Hilbert-space authorization. Gravity is structural necessity; quantum dynamics is a projection; their interaction is a joint-ledger problem.*

Theorem 7.3 (Gravity role precedes Hilbert projection). *Quantizing a metric projection cannot generate the gravity role that licensed the metric projection. A richer-scope quantum-gravity theory may add joint continuation structure, but it cannot serve as the primitive ground of gravity without projection backflow.*

Proof. The gravity role is forced at the physical-interior level as constraint architecture before metric field-equation form. The projective-Hilbert projection is another downstream representation of the same singleton interior. If Hilbert-space quantization is used to authorize gravity itself, the projection is being used to authorize the upstream role that licensed the projection. That is projection backflow. If additional joint structure is declared, the theory moves to richer-scope joint continuation rather than primitive same-scope generation of gravity. $\square$

8 Projection backflow and the error of quantum-gravity primacy

Definition 8.1 (Projection backflow). *Projection backflow occurs when a downstream projection is used to authorize, generate, repair, or select the upstream standing structure that licensed the projection, or when one faithful projection is used to authorize the other without a declared joint ledger.*

Theorem 8.2 (Projection backflow exclusion). *No faithful AASC projection may retro-authorize the standing of the physical interior or of another projection. A metric expression cannot authorize quantum standing; a quantum expression cannot authorize gravitational standing; a joint coupling cannot authorize either side unless it is declared as richer-scope joint continuation or a same-scope joint ledger.*

Proof. A projection is licensed after the physical interior has admissible standing. If a metric expression authorizes quantum standing, a downstream metric object has become a hidden classifier for a different projection. If a Hilbert expression authorizes gravitational standing, a downstream quantum object has become a hidden classifier for the gravity role. If a joint coupling authorizes either side without declaration, it performs boundary work after the act. All routes violate the fixed-domain consequence layer: no backflow, no second classifier, no same-act repair, and no undeclared carrier transfer. $\square$

Backflow route	Error
Quantizing the metric as if that authorizes gravity	Projective-Hilbert backflow into the already forced gravity role.
Treating the wavefunction as source of metric standing without a source ledger	Quantum-to-metric source laundering.
Treating curvature as primitive ground of quantum standing	Metric-to-quantum backflow.

Treating collapse as metric source update without record ledger	Update-to-source backflow and record laundering.
Treating entanglement as hidden metric transport	Compatibility-to-transport backflow.

Part III

Forced Projection Normal Forms

The paired normal forms below are not independent external premises. They are theorem-use summaries of the two faithful projection routes licensed by the same AASC physical-interior discipline. The Einstein theorem supplies the metric first-response projection of gravity as structural necessity; the Schrödinger theorem supplies the projective-Hilbert first-response projection of quantum standing.

9 Einstein dynamics theorem-use ladder

The companion Einstein dynamics theorem establishes the metric projection normal form. Its theorem-use form in this manuscript is:

$$\begin{aligned}
&\text{AASC gravity role} \Rightarrow \text{faithful metric projection} \\
&\Rightarrow \text{SameScopeMetric}_1(E_{\mu\nu}) \\
&\Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.
\end{aligned}$$

The result is a same-scope normal-form theorem, not a claim that every gravitational theory is Einstein gravity or that all constants are boundary-generated numbers [3, 2].

Theorem 9.1 (Einstein theorem-use package). *Within a faithful four-dimensional Lorentzian metric projection of the AASC gravity role, and under the minimal first-response same-scope predicate $\text{SameScopeMetric}_1(E_{\mu\nu})$, closed source continuation forces the metric equation form*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Remark 9.2 (Use in the present paper). *The present manuscript does not rederive the curvature-response exhaustion in full. It uses the Einstein theorem at theorem-use level: metric projection, first-response curvature normal form, divergence compatibility, source-ledger closure, and modified-gravity classification. The source paper provides the full audit expansion [3].*

10 Schrödinger dynamics theorem-use ladder

The companion Schrödinger dynamics theorem establishes the projective-Hilbert projection normal form. Its theorem-use form is:

$$\begin{aligned}
&\text{AASC quantum standing} \Rightarrow \text{faithful projective-Hilbert projection} \\
&\Rightarrow \text{SameScopeQuantumSpatial}_1(H) \\
&\Rightarrow i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.
\end{aligned}$$

The AASC-native normal form is $H = a\Delta + M_V$; the coefficient $a = -\hbar^2/(2m)$ is fixed after inertial/action scale anchoring.

Theorem 10.1 (Schrödinger theorem-use package). *Within a determinate nondegenerate closed autonomous scalar nonrelativistic projective-Hilbert projection of AASC quantum standing, and under SameScopeQuantumSpatial₁(H), the unique minimal first-response Hamiltonian form is*

$$H = a\Delta + M_V.$$

After admitted nonrelativistic inertial/action coefficient fixation, this gives

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

Remark 10.2 (Use in the present paper). *The present manuscript does not treat the wavefunction, Hilbert space, or Hamiltonian as primitive foundations. It uses the companion theorem at theorem-use level: projective state standing, Wigner-Stone closed continuation, Hamiltonian-response exhaustion, domain discipline, coefficient calibration, and modified-dynamics taxonomy. The provenance and operator-audit background are supplied by the AASC quantum-interior and operator papers [6, 7, 8, 9].*

11 Measurement-record construction theorem-use ladder

The admissible-record construction theorem supplies the record and certificate discipline needed for the joint-ledger layer. Its theorem-use form is:

$$\begin{aligned} &\text{physically non-selective quantum regime with record reuse} \\ &\Rightarrow \text{measurement-record quotient} \\ &\Rightarrow \text{fixation-commutation discipline.} \end{aligned}$$

Measurement records are fixed-domain constructed objects, not inert marks. Their evidential and metric-facing roles require admissible record-fixation, quotient-level standing, stable boundary-trace fixation where internal and provenance/indexical asymmetries fail, and downstream reuse discipline [10].

Theorem 11.1 (Measurement-record theorem-use package). *If a quantum theory is used to underwrite retained-record reuse, prospective-record reuse, deliberation, or evidential comparison, then admissible record-fixation assignments must either collapse into a single standing-bearing quotient class on the fixed domain or leave downstream use assignment-relative. Bare unitary dynamics, entanglement, decoherence, and branch-local record stability do not by themselves collapse the admissible class.*

Remark 11.2 (Use in the present paper). *The present manuscript uses the measurement theorem at theorem-use level. A quantum update, record, collapse, branch, or decoherence event becomes metric-facing only when the record-bearing role is fixed by an admissible record ledger. Without that ledger, a metric source change is record laundering or source laundering, not joint standing.*

12 Entanglement compatibility theorem-use ladder

The entanglement theorem supplies the no-hidden-transport discipline for the joint-ledger layer. Its theorem-use form is:

$$\text{Bell-scope witness reuse} \Rightarrow \text{joint standing class carrying tensor-level compatibility.}$$

On the fixed Bell scope, an entangled preparation is not two independently complete local standing states later connected by a mechanism. It is a joint standing class carrying tensor-level compatibility constraints; local measurement is admissible local realization, not tensor-changing transmission [11].

Theorem 12.1 (Entanglement theorem-use package). *Bell-violating, no-signaling correlations treated as invariant Bell-scope witness content are classified as boundary-level compatibility carried by a joint standing class. Hidden-variable, signaling, mechanism, delayed-repair, and extra-discriminator typings are not available as same-domain classifications.*

Remark 12.2 (Use in the present paper). *The present manuscript uses the entanglement theorem to block compatibility laundering. Entanglement may supply boundary-level compatibility inside the quantum projection, but it cannot be used as hidden metric communication, source transfer, record selection, horizon rescue, or same-scope standing repair without a declared joint ledger.*

13 Paired theorem-use ledger

The two theorem-use ladders share the same upstream roles and differ by projection.

Shared AASC role	Metric projection	Projective-Hilbert projection
Reference	Fixed spacetime/metric carrier and source target.	Fixed ray/projective state carrier and operator domain.
Standing	Closed metric/source continuation and tensorial redescription.	Closed transition-probability continuation and ray redescription.
Admissibility	Same-scope metric tensor discipline; no hidden curvature/source load.	Same-scope Hamiltonian/operator discipline; no hidden gauge/spin/update load.
Irreversibility	No deferred source, vacuum, horizon, or boundary repair.	No collapse, branch, normalization, or domain repair as same act.
Quotient identity	Diffeomorphism/lawful redescription; coordinates are skin.	Ray identity, basis covariance, global phase skin.
First response	Curvature obstruction.	Spatial Hamiltonian generator.
Normal form	$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$.	$i\hbar\partial_t\psi = (-\hbar^2\Delta/(2m) + M_V)\psi$.
Richer-scope routes	Higher curvature, torsion, non-locality, metric-affine structure.	Gauge, spin, relativistic, open, update, nonlocal, QFT structure.

Proposition 13.1 (Shared normal-form lesson). *Einstein dynamics and Schrödinger dynamics have the same proof shape inside AASC: each is a first-response same-scope projection normal form obtained only after target, carrier, lawful redescription, continuation, and no-hidden-load discipline are fixed.*

Part IV

Paired AASC Physical Interior and Same-Interior Fidelity

14 AASC physical interior with paired projection capacity

Definition 14.1 (Paired-projection physical interior). *A paired-projection AASC physical interior is a role-labelled structure*

$$\mathcal{I} = (\text{Anc}, \text{Ten}, \text{Skin}, \sim, \text{Cont}, \Pi_g, \Pi_H),$$

where *Anc* fixes the domain and admissibility envelope, *Ten* carries standing-bearing invariant content, *Skin* is representational surplus, $\sim$ is lawful identity-preserving redescription, *Cont* is admissible continuation, Π_g is a faithful metric projection, and Π_H is a faithful projective-Hilbert projection.

Definition 14.2 (Faithful paired projections). *The projections Π_g and Π_H are faithful iff:*

1. *each is downstream from the physical-interior target;*
2. *each preserves standing-bearing content up to its licensed presentation;*
3. *the kernel of each projection consists only of skin relative to its comparison class;*
4. *neither projection retro-authorizes the target or the other projection;*
5. *cross-projection influence occurs only through declared joint ledgers.*

15 Projection non-primacy theorem

Theorem 15.1 (Projection non-primacy). *In a paired-projection AASC physical interior,*

$$\Pi_g \not\Rightarrow \text{primitive authorization of } \Pi_H, \quad \Pi_H \not\Rightarrow \text{primitive authorization of } \Pi_g.$$

The metric cannot be the primitive ground of quantum standing, and the projective-Hilbert carrier cannot be the primitive ground of metric standing, within the same theorem act.

Proof. Both projections are licensed only after the physical-interior target has standing, reference, admissibility, irreversibility, quotient identity, and continuation. If a metric fact is used to authorize the quantum projection itself, a downstream metric tensor or geometry is made into an undeclared boundary selector for quantum standing. If a Hilbert fact is used to authorize the metric projection itself, a downstream representative state or operator is made into an undeclared boundary selector for metric standing. Both routes violate no-backflow discipline and confuse a theorem object with the admissibility interface that licensed it. Therefore neither projection is primitive over the other. $\square$

Corollary 15.2 (No primitive reduction). *A GR/QM structural-unification theorem cannot be a primitive reduction of quantum standing to metric geometry or metric standing to Hilbert geometry unless it declares a richer-scope target and supplies the missing admissibility, standing, redescription, and continuation ledgers. In the paired first-response target, both equations are downstream theorem objects.*

16 No projection backflow

Definition 16.1 (Projection backflow). *Projection backflow occurs when a downstream object from one faithful projection is used to authorize, repair, select, or reinterpret the standing of the other projection without a declared joint ledger.*

Theorem 16.2 (No projection backflow). *A metric object cannot be used as an undeclared quantum selector, and a quantum object cannot be used as an undeclared metric source. Cross-projection influence requires a declared joint ledger.*

Proof. A selector that changes standing-bearing use across projections performs admissibility-relevant work. If it is declared, it belongs to a joint ledger and changes the target from two independent projections to a joint continuation problem. If it is not declared, it is hidden selector work. Thus curvature cannot select a quantum branch, collapse cannot repair metric source closure, expectation values cannot source Einstein dynamics, and horizons cannot store quantum record standing without declared cross-ledger or certificate structure. $\square$

Backflow attempt	AASC status
Metric curvature selects a quantum branch.	Hidden measurement/update selector unless a record or collapse ledger is declared.
Quantum expectation value sources $G_{\mu\nu}$ directly.	Mixed projection requiring state, record, renormalization, and source-ledger closure.
Wavefunction collapse changes classical source as same act.	Measurement/update route; not closed metric/source continuation.
Horizon stores inaccessible quantum information.	Standing sink unless exterior certificate, leakage, non-sink quotient, or richer scope is declared.
Entanglement transmits metric influence.	Boundary compatibility is not interior transport; requires declared interaction or source ledger.

17 Boundary-trace fixation of paired roles

Definition 17.1 (Paired role-anchor fixation). *A paired role-anchor fixation is an admissibility-preserving assignment of role-anchor equivalence classes for metric content, source ledger, quantum state standing, operator domain, record structure, and cross-ledger coupling, without selecting primitive representatives.*

Proposition 17.2 (Representative-free paired fixation). *The metric role, source role, quantum state role, operator-domain role, record role, and cross-ledger role must be fixed as equivalence classes, not primitive representatives. Coordinates, frames, bases, global phases, branch labels, and index labels cannot carry primitive paired standing.*

Proof. Each listed role participates in standing-bearing continuation. If one were fixed by representative privilege, the joint theorem would import a hidden selector. Boundary-trace or declared role-anchor fixation supplies the needed role class without selecting a coordinate, basis, branch, or record representative as primitive. $\square$

Part V

Joint Ledger as Cross-Projection Closure

18 Joint metric/quantum ledger

The paired normal-form theorem shows why Einstein and Schrödinger dynamics are not primitive rivals. Coupling them requires more than writing one equation beside the other. It requires a joint ledger that tracks what each projection needs in order to preserve standing and what every cross-projection term is allowed to do.

Definition 18.1 (Expanded joint metric/quantum ledger). *A joint metric/quantum ledger is a tuple*

$$\mathcal{L}_{gH} = (\mathcal{L}_g, \mathcal{L}_H, \mathcal{L}_{\text{cross}}, \mathcal{L}_{\text{rec}}, \mathcal{L}_{\text{comp}}, \mathcal{C}_{\text{cert}}, \text{Cont}_{\text{joint}}, \text{Rep}),$$

where:

1. $\mathcal{L}_g$ is the metric/source ledger;
2. $\mathcal{L}_H$ is the quantum transition ledger;
3. $\mathcal{L}_{\text{cross}}$ is the declared metric/quantum cross-response ledger;
4. $\mathcal{L}_{\text{rec}}$ is the admissible record-fixation ledger;
5. $\mathcal{L}_{\text{comp}}$ is the boundary-compatibility and entanglement ledger;
6. $\mathcal{C}_{\text{cert}}$ is the certificate family for records, sources, boundaries, horizons, renormalization, and exterior reconstruction;
7. $\text{Cont}_{\text{joint}}$ is the admitted joint continuation relation; and
8. Rep is the reportability and claim-standing ledger.

Definition 18.2 (Metric source ledger). *The metric source ledger $\mathcal{L}_g$ contains the standing-bearing deformation-response, field-carrier, vacuum-baseline, boundary, exchange, and closure content required for a closed or declared-open metric source tensor $T_{\mu\nu}^{\text{joint}}$. In closed scope it supplies*

$$\nabla^\mu T_{\mu\nu}^{\text{joint}} = 0.$$

In open or effective scope it declares the exchange term

$$\nabla^\mu T_{\mu\nu}^{\text{joint}} = J_\nu^{\text{declared}}.$$

Definition 18.3 (Quantum transition ledger). *The quantum transition ledger $\mathcal{L}_H$ contains projective state identity, transition-probability standing, operator-domain status, evolution/update type, closed/open classification, and any quantum field, state, algebra, or density-operator content used by the joint claim.*

Definition 18.4 (Record-fixation ledger). *The record-fixation ledger is the measurement-facing subledger*

$$\mathcal{L}_{\text{rec}} = (\text{Rec}, A_{\text{Ref}}/\approx_{\text{MR}}, \text{FixComm}, \text{Trace}, \text{Reuse}, \text{Cert}_{\text{rec}}, \text{Report}),$$

where Rec is the record-bearing structure; $A_{\text{Ref}}/\approx_{\text{MR}}$ is the admissible measurement-record quotient class; FixComm records fixation-commutation under lawful same-domain redescription; Trace is the declared singleton boundary trace when internal and provenance/indexical asymmetries do not fix the role; Reuse is retained-record, prospective-record, deliberative, or evidential downstream reuse; Cert_{rec} certifies record role, lineage, and reportable scope; and Report is the reportable record output. This ledger is required whenever a quantum event, measurement, branch, decoherence structure, update, or record claim is used as metric-facing source content.

Proposition 18.5 (Measurement-record theorem-use ladder). *The record ledger imports the measurement-record theorem only at the following theorem-use points: admissible record-fixation assignments are forced by fixed-domain downstream reuse; standing-bearing record content factors through the measurement-record quotient $A_{\text{Ref}}/\approx_{\text{MR}}$; evidential and source-relevant use requires fixation commutation; after internal and provenance/indexical asymmetries are exhausted, singleton boundary-trace fixation is the residual admissible same-domain route; and no fifth selector locus remains. Thus a measurement or update event cannot become metric-facing by record rhetoric alone.*

Proof. The measurement-record theorem treats records as fixed-domain constructed objects, not inert marks. If a record is used downstream, the use must factor through the admissible quotient and must preserve its role under lawful redescription. If the record role is not fixed, the record does not supply a construction-relative source or reportable evidential object. The entries of $\mathcal{L}_{\text{rec}}$ are exactly the roles required by that theorem-use ladder; omitting any one either leaves the record assignment-relative, imports a selector, changes scope, or fails standing. $\square$

Definition 18.6 (Boundary-compatibility ledger). *The compatibility ledger $\mathcal{L}_{\text{comp}}$ contains joint standing-class compatibility content, including entanglement/Bell-scope compatibility when relevant. Its theorem-use ladder is:*

$$\begin{aligned} &\text{Bell witness fixation} \Rightarrow \text{joint standing class} \Rightarrow \text{tensor compatibility,} \\ &\text{no hidden-variable/signaling/mechanism/repair typing} \Rightarrow \text{boundary compatibility.} \end{aligned}$$

It is not an interior transport channel. If it moves, repairs, or transfers standing-bearing content, it has left compatibility status and must be retyped as declared tensor/operator load, cross-response ledger, boundary certificate, open/effective scope, richer scope, or hidden repair.

Proposition 18.7 (Compatibility theorem-use ladder). *On a Bell or entanglement-relevant joint target, compatibility standing is carried by a joint standing class and sampled by admissible local realizations. It cannot be used as a hidden metric communication channel, source transfer, horizon rescue, branch selector, or same-act repair. A model that needs those roles must declare them in $\mathcal{L}_{\text{cross}}$, $\mathcal{C}_{\text{cert}}$, open/effective scope, or richer joint continuation.*

Proof. The entanglement theorem classifies Bell correlations as tensor-level compatibility of a joint standing class. It closes hidden-variable, signaling, mechanism, delayed-repair, and extra-discriminator typings on the same Bell scope. Therefore a cross-projection model may use entanglement as compatibility content, but not as undeclared transport or source motion. If the model uses compatibility to move standing-bearing content, the role has changed and must be declared outside pure compatibility. $\square$

Definition 18.8 (Cross-response ledger). *The cross-response ledger $\mathcal{L}_{\text{cross}}$ contains all declared roles by which quantum content affects metric source standing or metric content affects quantum continuation standing. It must state whether the effect is closed, open, effective, measurement/update, semiclassical, field-theoretic, boundary/certificate, or richer-scope.*

19 Joint compatibility conditions

Definition 19.1 (Joint-ledger admissibility predicate). *A joint ledger $\mathcal{L}_{gH}$ is admissible iff it satisfies:*

1. **Metric closure or declared exchange:** $\nabla^\mu T_{\mu\nu}^{\text{joint}} = 0$ in closed scope, or every nonzero divergence is assigned to a declared exchange, anomaly, boundary, open, or effective ledger.
2. **Quantum closure or declared update:** transition-standing is preserved in closed scope, or every failure is assigned to a declared measurement, record, bath, stochastic, open, or effective ledger.
3. **Record certification:** measurement, update, branch, decoherence, or outcome content used as metric-facing source content has an admissible $\mathcal{L}_{\text{rec}}$ ledger.
4. **Compatibility discipline:** entanglement and boundary compatibility are not retyped as hidden metric transport, signal, or standing repair.
5. **Lawful redescription on both projections:** diffeomorphism and projective-Hilbert redescrptions preserve standing-bearing content.
6. **No hidden branch selector:** branch, collapse, record, or outcome selection is declared if it changes standing.
7. **No hidden source import:** expectation values, fluctuations, renormalized tensors, and effective sources are declared with closure conditions.
8. **No same-act repair:** later records, boundary certificates, renormalization choices, or branch narratives cannot repair failed earlier standing as the same act.
9. **Reportable scope:** the claim reports only the scope carried by the ledger.

Theorem 19.2 (Joint-ledger adequacy theorem). *Any GR/QM structural-unification claim must supply a joint ledger satisfying the joint-ledger admissibility predicate. If it does not, it is not an admissible cross-projection interaction of the paired physical interior; it is a background approximation, effective model, mixed-scope construction, hidden selector, source laundering, record laundering, compatibility laundering, or standing failure.*

Proof. A unification claim asserts that metric and quantum contents participate in one standing-bearing physical continuation. The metric projection requires source closure or declared exchange. The quantum projection requires transition-standing closure or declared update/open scope. Measurement and record claims require admissible record fixation. Entanglement and nonlocal compatibility require compatibility discipline rather than hidden transport. Cross-effects change standing in at least one projection and therefore require admissibility-bearing ledger status. If no such ledger is supplied, the claim either uses one projection as a hidden selector for the other, omits source/record exchange, repairs a failed act later, or changes scope without declaration. These are exactly the excluded cases. $\square$

20 Same-scope joint-coupling candidates

The joint ledger is the survivor of the same-scope cross-projection target, just as the Einstein tensor form and the Schrödinger Hamiltonian form are survivors of their respective first-response predicates.

The predicate below is narrow because the target is narrow: a broader predicate does not refute this theorem; it defines a different theorem object.

Definition 20.1 (Same-scope first-response joint-coupling predicate). *Let C be a proposed cross-projection coupling between the metric and projective-Hilbert projections of the singleton interior. Write $\text{SameScopeJoint}_1(C)$ iff all of the following hold:*

1. C is internal to the same singleton physical-interior target and does not split $\mathcal{I}$ into $\mathcal{I}_g$ and $\mathcal{I}_H$.
2. C preserves projection non-primacy: neither projection authorizes, generates, or repairs the other.
3. Any metric-standing effect of C is carried by $\mathcal{L}_g$ as closed source content or declared exchange.
4. Any quantum-standing effect of C is carried by $\mathcal{L}_H$ as closed transition standing or declared update/open status.
5. Any measurement, branch, record, collapse, or decoherence content used by C is certified by $\mathcal{L}_{\text{rec}}$.
6. Any entanglement, Bell, boundary, or nonlocal compatibility content used by C is certified by $\mathcal{L}_{\text{comp}}$ and is not retyped as transport.
7. Any boundary, horizon, renormalization, apparatus, exterior-reconstruction, or regulator role used by C is declared in $\mathcal{C}_{\text{cert}}$.
8. C is lawful under both metric redescription and projective-Hilbert/ray redescription.
9. C contains no hidden branch selector, hidden source import, hidden record selector, hidden compatibility transport, or hidden scope transfer.
10. C reports only the scope carried by Rep.

Predicate clause weakened	Cost
Same singleton interior	The claim becomes external coordination of two domains or hidden plural-interior split.
No projection backflow	A downstream projection authorizes upstream standing.
Metric ledger	Source laundering or metric source-standing failure.
Quantum ledger	Hidden update, open/effective scope, or quantum standing failure.
Record ledger	Measurement/update becomes record laundering.
Compatibility ledger	Entanglement becomes hidden transport, source transfer, or repair.
Certificate ledger	Boundary, horizon, renormalization, apparatus, or exterior claim becomes uncertified.
Lawful redescription	Coordinate, basis, branch, or representative privilege carries force.
No hidden selectors	Branch, record, source, or horizon choice is imported without standing.

Proposition 20.2 (Predicate-target correspondence). *SameScopeJoint₁(C) is not a definition chosen to force a desired conclusion. Each clause is the cross-projection translation of a target burden already present in the singleton physical-interior theorem: one interior, no backflow, metric source closure, quantum transition standing, record fixation, boundary compatibility, certificate discipline, lawful redescription, no same-act repair, and reportability.*

21 No source, record, or compatibility laundering

Definition 21.1 (Source laundering). *Source laundering occurs when a quantity from a quantum representation is inserted into a metric source role without the state, record, renormalization, closure, and continuation certificates required for metric standing.*

Definition 21.2 (Record laundering). *Record laundering occurs when a measurement, branch, decoherence, or outcome structure is invoked as quantum standing or metric-facing source content without declaring its admissible record-fixation ledger.*

Definition 21.3 (Compatibility laundering). *Compatibility laundering occurs when entanglement or boundary-level compatibility is treated as hidden metric transport, source transfer, superluminal communication, or same-scope standing repair.*

Proposition 21.4 (Laundering exclusion). *A joint ledger blocks source laundering, record laundering, and compatibility laundering by forcing every cross-projection standing change to appear as declared tensor/operator content, declared record/update, declared effective/open scope, declared boundary certificate, or richer-scope target.*

Part VI

Measurement, Records, Entanglement, and Joint-Ledger Certification

22 Measurement-to-source certification

Theorem 22.1 (Measurement-to-source certification). *A measurement, update, branch-selection, decoherence, or observer-relative event may enter a metric-facing source ledger only if its record-bearing role is fixed by an admissible record ledger*

$$\mathcal{L}_{\text{rec}} = (\text{Rec}, A_{\text{Ref}}/\approx_{MR}, \text{FixComm}, \text{Trace}, \text{Reuse}, \text{Cert}_{\text{rec}}, \text{Report}).$$

If a quantum event changes metric-facing source content without such a ledger, then the move is hidden record selection, source laundering, collapse/update import as same-scope repair, open/effective scope, or standing failure.

Proof. The admissible-record theorem classifies measurement records as fixed-domain constructed objects whose standing-bearing use requires record-fixation quotient discipline, fixation commutation,

boundary-trace/certificate discipline, and downstream reuse. A metric source change is standing-bearing metric content. If the source change depends on a measurement or update event, then the event must have a fixed record-bearing role. Without $\mathcal{L}_{\text{rec}}$, the event does not supply a determinate source-facing record. Treating it as source content nevertheless either imports a selector, inserts an uncertified source, repairs an earlier failure, or changes to an open/effective target. These cases exhaust the non-ledger alternatives. $\square$

Corollary 22.2 (No uncertified expectation-value sourcing). *An expression such as $\langle \hat{T}_{\mu\nu} \rangle$ does not by itself define a closed metric source. It becomes metric-facing only after state, renormalization, record/update, cross-response, and source-closure ledgers are fixed.*

23 No uncertified collapse or update sourcing

Theorem 23.1 (No uncertified collapse/update sourcing). *A collapse, projection, branch-selection, decoherence event, or observer-relative update cannot modify the same-scope metric source ledger unless it is declared as dynamic selection, ontic privileging, record-fixation restriction to one quotient class, constructional deflation/open scope, or richer-scope theory. Otherwise the move is source laundering or same-act repair.*

Proof. The measurement-record theorem exhausts the loci of singular record-fixation work: dynamic selection, ontic privileging, constructional restriction, or constructional deflation. If an update changes metric source standing, it must occupy one of those declared loci or change scope. If it occupies none, it supplies a metric-facing consequence without a fixed record role, and so it is either hidden selector work, uncertified source import, or later repair. $\square$

24 Boundary compatibility is not metric transport

Theorem 24.1 (Boundary compatibility is not metric transport). *An entangled preparation may supply joint tensor compatibility in the quantum projection, but that compatibility cannot be used as same-scope metric transport, source transfer, superluminal communication, or hidden repair. Symbolically,*

$$\mathcal{L}_{\text{comp}} \neq \text{interior transport}.$$

If compatibility is used to move standing-bearing content, it must be retyped as declared tensor/operator load, declared cross-ledger, boundary-trace certificate, open/effective scope, richer-scope theory, hidden repair, or standing failure.

Proof. The entanglement theorem types Bell-scope correlations as tensor-level compatibility carried by a joint standing class. Local measurement is admissible local realization of that compatibility, not tensor-changing transmission. Metric transport, communication, or source transfer would be a different role: it would move or repair standing-bearing content inside the admitted metric projection. That role is not supplied by boundary compatibility. If a model intends compatibility to do transport work, the transport must be declared as a cross-ledger term or richer scope. If it is undeclared, it is compatibility laundering. $\square$

25 Record-certified source extraction

Definition 25.1 (Record-certified source extraction). *A metric-facing quantum source extraction is a map*

$$T_{\mu\nu}^{q,\text{cert}} = \text{Extract}_g(\mathcal{L}_H, \mathcal{L}_{\text{rec}}, \mathcal{L}_{\text{cross}}, \mathcal{C}_{\text{cert}}),$$

where $\mathcal{L}_H$ fixes the quantum state/ray/field content, $\mathcal{L}_{\text{rec}}$ fixes any measurement/update/record lineage used by the source claim, $\mathcal{L}_{\text{cross}}$ declares how the quantum content becomes metric-facing, and $\mathcal{C}_{\text{cert}}$ certifies renormalization, domain, boundary, and closure status.

Theorem 25.2 (Record-certified source closure). *A record-certified quantum source is closed in the metric projection only when*

$$\nabla^\mu T_{\mu\nu}^{q,\text{cert}} = 0.$$

If the divergence is nonzero, the model must declare an exchange ledger

$$\nabla^\mu T_{\mu\nu}^{q,\text{cert}} = J_\nu^{\text{declared}}.$$

Without closure or declared exchange, the source extraction is incomplete and cannot bear same-scope metric source standing.

26 Cost of joint-ledger denial

Denied ledger element	Joint GR/QM cost
$\mathcal{L}_g$	No metric source closure; the Einstein side loses theorem-bearing source status.
$\mathcal{L}_H$	No quantum transition-standing, state identity, operator-domain status, or update classification.
$\mathcal{L}_{\text{rec}}$	Measurement/update cannot become reportable source or evidence; record laundering enters.
$\mathcal{L}_{\text{comp}}$	Entanglement is misread as transport, hidden mechanism, signaling, or source repair.
$\mathcal{L}_{\text{cross}}$	No coupling is declared; only parallel projections remain.
$\mathcal{C}_{\text{cert}}$	Record, source, renormalization, boundary, horizon, or exterior claims become uncertified.
$\text{Cont}_{\text{joint}}$	No unified continuation; only mixed-scope fragments remain.
Rep	An approximation, presentation, or effective model is overreported as unification.

Part VII

Worked Audits of Existing GR/QM Interface Routes

27 Worked Audit I: semiclassical gravity

The semiclassical Einstein equation has the schematic form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}}.$$

It is not rejected. It is typed as an effective mixed-projection ledger whose standing depends on state, record, renormalization, and source-closure certification.

Ledger component	Required content in semiclassical gravity
$\mathcal{L}_g$	Metric carrier, Einstein-side source target, divergence closure or declared exchange.
$\mathcal{L}_H$	Quantum state ω , field algebra, domain, state class, and update/open status.
$\mathcal{L}_{\text{cross}}$	Expectation-value extraction route from quantum field content to metric-facing tensor.
$\mathcal{L}_{\text{rec}}$	Record/update status if ω depends on measurement, collapse, branch, or decoherence.
$\mathcal{C}_{\text{cert}}$	Renormalization prescription, state certificate, domain certificate, closure certificate.
$\text{Cont}_{\text{joint}}$	Joint evolution if claimed, or effective/mixed-scope status if not.

Proposition 27.1 (Semiclassical pass/fail audit). *A semiclassical source extraction must pass the following ledger questions before it can stand as a metric-facing source:*

1. What fixes the algebraic or Hilbert state ω ?
2. What fixes the field algebra, domain, and state class?
3. What certifies the renormalized tensor $\langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}}$?
4. Does $\nabla^{\mu} \langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}} = 0$ hold in the declared metric projection?
5. If the divergence is nonzero, what exchange ledger $J_{\nu}^{\text{declared}}$ is declared?
6. Does ω depend on measurement, collapse, branch, decoherence, conditioning, or record update?
7. If so, where is the expanded record ledger $\mathcal{L}_{\text{rec}}$?
8. Is backreaction a closed joint continuation, an open/effective correction, or only a background approximation?

Failure at any point does not refute semiclassical gravity; it fixes its reportable status as incomplete, effective, open, or source-laundered relative to the structural-unification target.

Theorem 27.2 (Semiclassical gravity audit). *The semiclassical Einstein equation is admissible as an effective mixed-projection ledger only if $\langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}}$ is state-certified, renormalization-certified, record-certified when measurement/update enters, and divergence-compatible or declared open. Otherwise it is source laundering.*

Proof. The left side is an Einstein-side metric response requiring a closed or declared-open source. The right side is extracted from quantum field content. If the state, renormalization prescription, record/update status, or divergence behavior is not declared, the expression does not provide the metric source ledger required by the Einstein theorem-use package. If all are declared, the equation may stand as an effective mixed-projection ledger, but not as a completed joint dynamics unless a joint continuation is also supplied. $\square$

Corollary 27.3 (Semiclassical effective-ledger status). *The equation*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}}$$

has AASC standing as an effective mixed-projection ledger iff the state ω , field algebra/domain, renormalization prescription, record/update status when relevant, and divergence closure or declared exchange are fixed. If any of these are used as standing-bearing content while undeclared, the equation is source laundering, record laundering, or open/effective scope without reportable certification.

28 Worked Audit II: QFT on curved spacetime

QFT on curved spacetime uses a fixed or externally controlled metric as background anchor and quantizes fields over that anchor.

Proposition 28.1 (QFT-on-curved-spacetime classification). *QFT on curved spacetime is a projective-Hilbert/QFT enrichment over a metric anchor. It is not full joint continuation unless backreaction and metric source closure are promoted into a declared joint ledger.*

Proof. The metric is not produced by the quantum theory in the basic QFT-on-curved-spacetime setting; it functions as a background anchor. Quantum fields carry Hilbert/Fock/algebraic structure over that anchor. Renormalized stress tensors may enter effective source ledgers, but only under the semiclassical certification conditions above. Therefore the theory is a powerful mixed-scope framework, not a same-scope cross-projection theorem by itself. $\square$

29 Worked Audit III: canonical and Wheeler-DeWitt routes

Canonical quantum gravity attempts to make the gravitational degrees of freedom themselves quantum. Its AASC classification depends on whether the construction preserves metric source standing and quantum transition standing in one declared continuation.

Proposition 29.1 (Canonical quantum-gravity classification). *A canonical or Wheeler-DeWitt route is admissible as richer-scope quantum-gravity structure only if it supplies the role of time, constraint closure, record standing, source extraction, and reportability. If it loses continuation standing or imports a hidden clock/branch selector, it has not supplied the joint ledger required here.*

30 Worked Audit IV: path-integral and histories routes

Path-integral quantum gravity and histories programs use sums, amplitudes, or measures over histories or geometries.

Proposition 30.1 (Path-integral audit). *A path-integral or histories route becomes an admissible joint-ledger candidate only when its measure, regulator, boundary data, gauge fixing, record/certificate structure, and reportable target are declared. Otherwise the measure or boundary choice functions as hidden admissibility authority.*

31 Worked Audit V: collapse, stochastic gravity, and metric source change

Theorem 31.1 (Collapse/stochastic gravity audit). *Collapse or stochastic update is not same-scope closed Schrödinger continuation. It is a declared update/dynamics enrichment. It may source metric content only if its record/source ledger is declared and metric divergence compatibility is preserved or exchange is declared.*

Proof. A collapse or stochastic update changes the quantum ledger by construction. If it also changes metric source content, the change must pass through $\mathcal{L}_{\text{rec}}$, $\mathcal{L}_{\text{cross}}$, and $\mathcal{L}_g$. Without those ledgers, the update is treated as source content without record or closure certification. With those ledgers, the model is a declared richer/open/update theory and may be evaluated at that scope. $\square$

32 Worked Audit VI: entanglement, horizons, and nonlocal compatibility

Theorem 32.1 (Entanglement/horizon route closure). *A horizon or nonlocal GR/QM candidate cannot preserve standing by appealing to entanglement alone. Boundary-level compatibility cannot function as hidden metric transport, horizon information sink, source-repair channel, nonlocal same-scope communication, or measurement-record selector. The model must provide exterior witness algebra, leakage/radiation channel, boundary-to-boundary certificate, non-sink quotient, or declared richer-scope tensor load.*

Proof. Entanglement supplies compatibility, not transport. A horizon threatens standing when it blocks record, source, or exterior reuse. If the threatened content is standing-bearing, compatibility alone does not move it across the metric projection or repair exterior standing. A certificate, leakage channel, non-sink quotient, or richer-scope ledger is required. If none is supplied, the horizon or nonlocal route is compatibility laundering or standing failure. $\square$

33 Interface taxonomy table

Interface route	AASC joint-ledger classification
Semiclassical Einstein equation	Effective mixed-projection ledger if state, renormalization, record, and source closure are certified; otherwise source laundering.

QFT on curved spacetime	Quantum field projection over metric anchor; backreaction requires joint ledger.
Canonical/Wheeler-DeWitt	Richer-scope quantum-gravity candidate; must solve constraint/time/record/source reportability.
Path integral / histories	Measure, regulator, boundary, record, and target certificates required.
Emergent spacetime / entanglement geometry	Hilbert-to-metric reconstruction route; must avoid projection backflow and compatibility laundering.
Objective collapse plus gravity	Declared update/dynamic-selection route; metric source closure still required.
Many-worlds branching	Branch/record ledger required; branch multiplicity alone does not fix metric-facing source standing.
Decoherence	Environment/record ledger; not hidden outcome or source selection.
Holography / boundary reconstruction	Boundary/certificate route when exterior witness algebra and reconstruction map are declared.
Horizon information claims	Standing sink unless leakage, certificate, non-sink quotient, or richer scope is supplied.

Part VIII

Compatibility Squares and Same-Scope Cross-Coupling Exhaustion

34 Compatibility-square definitions

Definition 34.1 (Metric compatibility square). *The metric square commutes when*

$$\Pi_g \circ \text{Cont}_{\mathcal{I}} \simeq \text{Cont}_g \circ \Pi_g$$

up to declared metric exchange terms. Failure without declaration is metric standing failure or source laundering.

Definition 34.2 (Quantum compatibility square). *The quantum square commutes when*

$$\Pi_H \circ \text{Cont}_{\mathcal{I}} \simeq \text{Cont}_H \circ \Pi_H$$

up to declared update, open, measurement, or effective terms. Failure without declaration is quantum transition-standing failure or update laundering.

Definition 34.3 (Record compatibility square). *The record square commutes when*

$$\Pi_{\text{rec}} \circ \text{Cont}_{\mathcal{I}} \simeq \text{Cont}_{\text{rec}} \circ \Pi_{\text{rec}}$$

up to declared record, apparatus, environment, boundary-trace, or reportability certificates. Failure without declaration is record laundering.

Definition 34.4 (Joint compatibility square). *The joint square commutes when*

$$\Pi_{gH} \circ \text{Cont}_{\mathcal{I}} \simeq \text{Cont}_{\text{joint}} \circ \Pi_{gH}$$

where the equivalence is witnessed by $\mathcal{L}_{gH}$. Failure without declaration is projection backflow, hidden coupling, or standing failure.

35 Compatibility-square failure theorem

Theorem 35.1 (Compatibility-square failure theorem). *If the metric square, quantum square, record square, or joint square fails, at least one of the following typed routes must be declared: cross-ledger exchange, open/effective scope, measurement/update regime, boundary/certificate route, richer-scope projection, projection backflow, hidden selector, or standing failure.*

Proof. A square failure means that projection after interior continuation does not match continuation after projection. If the mismatch is standing-bearing and declared, it belongs to a cross-ledger exchange, open/effective scope, measurement/update regime, boundary/certificate route, or richer-scope projection. If the mismatch is standing-bearing and undeclared, it either uses one projection to authorize the other, imports a hidden selector, or leaves standing unsupported. If it is not standing-bearing, it is skin. These cases exhaust the fixed-domain options because A^+ excludes a second same-scope classifier and same-act repair. $\square$

36 The central joint predicate

This structural-unification paper has its own same-scope exhaustion predicate. It plays the same role in the cross-projection layer that the metric and Hamiltonian predicates play in the two normal-form papers:

Theorem	Predicate	Survivor
Einstein	$\text{SameScopeMetric}_1(E_{\mu\nu})$	Einstein metric response form
Schrödinger	$\text{SameScopeQuantumSpatial}_1(H)$	$H = a\Delta + M_V$
Structural unification	$\text{SameScopeJoint}_1(C)$	$C \in \mathcal{L}_{gH}$

The predicate is narrow because the cross-projection target is narrow. A broader predicate does not refute the theorem; it defines a different theorem object. The burden of the present theorem is to exhaust same-target, same-interior, first cross-projection standing-bearing couplings without projection backflow, source laundering, record laundering, compatibility laundering, hidden certificates, or reportability inflation.

37 Same-scope joint coupling candidates

Definition 37.1 (Same-scope joint coupling candidate). *A same-scope joint coupling candidate C is any proposed same-target rule, term, equation, transition, source-extraction operation, record/update claim, compatibility claim, boundary/certificate claim, or reconstruction route that changes or purports to change standing-bearing content across the metric and projective-Hilbert projections while claiming to preserve the paired physical-interior target. When useful for audit, write*

$$C = (C_g, C_H, C_{\text{cross}}, C_{\text{rec}}, C_{\text{comp}}, C_{\text{cert}}, C_{\text{cont}}, C_{\text{rep}}),$$

where each coordinate may be empty, inert, declared, or standing-bearing.

Definition 37.2 (The predicate $\text{SameScopeJoint}_1(C)$). Write $\text{SameScopeJoint}_1(C)$ when C satisfies all of the following:

1. same physical-interior target and same-interior fidelity;
2. no projection backflow;
3. metric source closure or declared exchange;
4. quantum transition closure or declared measurement/open/update status;
5. record-certified measurement/update use whenever records, branches, outcomes, or decoherence enter the coupling;
6. compatibility is not treated as transport, source transfer, or hidden repair;
7. boundary, horizon, renormalization, regulator, and apparatus certificates are declared when they carry standing;
8. lawful redescription is preserved in both projections;
9. no hidden branch, horizon, record, or renormalization selector is imported;
10. reportable-scope discipline is fixed.

Proposition 37.3 (Cost of weakening SameScopeJoint_1). Weakening any load-bearing clause of $\text{SameScopeJoint}_1(C)$ changes the theorem target rather than producing an admissible same-scope counterexample.

<i>Clause weakened</i>	<i>Cost</i>
<i>Same physical-interior target</i>	<i>The coupling splits the singleton interior or changes scope.</i>
<i>No projection backflow</i>	<i>One projection is used to authorize the upstream standing of the other.</i>
<i>Metric source closure</i>	<i>Metric-side source standing fails or source laundering enters.</i>
<i>Quantum transition/update status</i>	<i>Quantum standing fails, or hidden update/open scope enters.</i>
<i>Record certification</i>	<i>Measurement, branch, outcome, or decoherence content is record-laundered.</i>
<i>Compatibility discipline</i>	<i>Entanglement is misread as transport, source transfer, or hidden repair.</i>
<i>Certificate declaration</i>	<i>Boundary, horizon, renormalization, or apparatus claims become uncertified selectors.</i>
<i>Lawful redescription</i>	<i>Coordinates, basis, branch labels, or presentation skin acquire hidden force.</i>
<i>Reportability</i>	<i>Effective or mixed-scope material is overreported as structural unification.</i>

Part IX

No Primitive Conflict and Quantum-Gravity Retyping

38 Diagnosis of the apparent conflict

The usual framing says that GR is geometric, classical, and diffeomorphism-invariant, while QM is Hilbertian, probabilistic, and unitary; therefore one formalism must be reduced to the other. AASC changes the target. GR and QM are not primitive rivals at the kernel level. Each is a projection normal form of the physical interior.

The conflict appears when:

1. the metric projection is treated as primitive over quantum standing;
2. the Hilbert projection is treated as primitive over metric standing;
3. expectation values are used as sources without closure and record ledgers;
4. measurement/update is treated as closed evolution;
5. background anchors are reported as generated theorem content;
6. projections are crossed without declaring the cross-ledger.

39 No primitive conflict theorem

Theorem 39.1 (No primitive conflict). *If Einstein dynamics and Schrödinger dynamics are treated as faithful first-response projections of the same AASC physical interior, then they are not primitive rivals. Any genuine incompatibility arises at the joint-ledger level, not at the kernel or first-response projection-normal-form level.*

Proof. The kernel roles required by both projections are identical: admissibility, standing, reference, and irreversibility. The metric and projective-Hilbert carriers are downstream projection envelopes. Each first-response theorem exhausts its own same-scope candidates and classifies richer alternatives without denying them. Therefore no contradiction exists merely because the two projections use different carriers, redescription laws, or first-response objects. A conflict appears only when one attempts to couple them: metric source closure and quantum transition standing must then be preserved by one joint ledger. Failure there is a joint-ledger failure, not a primitive kernel conflict. $\square$

Corollary 39.2 (Unification reframing). *GR/QM unification is not the task of forcing one first-response projection to be primitive over the other. It is the task of constructing or classifying joint ledgers that preserve both projection continuations without hidden selectors, source laundering, record laundering, or same-act repair.*

Part X

Minimal Joint-Projection Compatibility

40 Compatibility diagram

Let

$$\mathcal{I} \xrightarrow{\Pi_g} (M, g_{\mu\nu}, T_{\mu\nu}^{\text{closed}})$$

and

$$\mathcal{I} \xrightarrow{\Pi_H} (\mathcal{H}, \mathbb{P}(\mathcal{H}), H, \text{TP}).$$

A joint continuation should make the projection diagrams commute up to declared ledger terms:

$$\Pi_g \circ \text{Cont}_{\mathcal{I}} \simeq \text{Cont}_g \circ \Pi_g, \quad \Pi_H \circ \text{Cont}_{\mathcal{I}} \simeq \text{Cont}_H \circ \Pi_H.$$

The symbol $\simeq$ means equality of standing-bearing output modulo lawful redescription, skin, and declared joint-ledger exchange. It does not allow hidden repair.

Definition 40.1 (Compatibility square). *A projection square commutes admissibly iff the upstream continuation and the projected continuation produce the same standing-bearing output up to lawful redescription and declared exchange ledgers.*

Theorem 40.2 (Compatibility square theorem). *A joint GR/QM continuation is admissible only if both the metric and projective-Hilbert compatibility squares commute admissibly. If either square fails without declared exchange, the model has hidden repair, source/record laundering, or standing failure.*

Proof. The projection of a standing-preserving upstream continuation must preserve standing in each faithful projection. If the metric square fails, the metric source or geometric response has changed without a declared source/exchange ledger. If the Hilbert square fails, quantum transition standing or update status has changed without a declared record/open/update ledger. Either failure violates admissibility unless declared as part of the joint target. $\square$

41 Joint closure theorem

Definition 41.1 (Joint closure). *A joint ledger is closed iff it satisfies metric closure*

$$\nabla^\mu T_{\mu\nu}^{\text{joint}} = 0$$

in the declared metric projection and quantum transition-standing closure in the declared projective-Hilbert projection, with all cross-exchanges included in $\mathcal{L}_{\text{cross}}$.

Theorem 41.2 (Joint closure necessity). *Every closed GR/QM structural-unification theorem must provide metric divergence closure, quantum transition-standing closure, and a declared cross-ledger. If any term is omitted, the target is open/effective, mixed-scope, or standing-failed rather than closed unification.*

Proof. Metric closure is required by the Einstein projection theorem. Quantum transition-standing closure is required by the Schrödinger projection theorem in closed autonomous scope. A coupling changes at least one projection's standing-bearing content. Therefore the coupling must be recorded in a cross-ledger that preserves or declares the relevant exchange. Without it, the model is using hidden source or hidden update authority. $\square$

42 Minimal joint-ledger predicate

Definition 42.1 (Minimal joint-ledger predicate). *Write $\text{JointLedger}_1(\mathcal{L}_{gH})$ when the joint ledger satisfies:*

1. *fixed physical-interior target;*
2. *faithful metric and projective-Hilbert projections;*
3. *no projection backflow;*
4. *metric closure or declared exchange;*
5. *quantum transition closure or declared update/open scope;*
6. *cross-response ledger with source, record, and boundary certificates;*
7. *lawful redescription on both sides;*
8. *no hidden branch, horizon, or measurement selector;*
9. *no undeclared renormalization or regulator authority;*
10. *reportable-scope discipline.*

Proposition 42.2 (Cost of weakening joint ledger). *Removing any clause from JointLedger_1 either destroys the paired target or moves the candidate into a declared richer/effective/open/update scope. A broader predicate does not refute the theorem; it defines a different theorem object.*

Clause weakened	Cost
No projection backflow	Metric or Hilbert presentation becomes hidden selector.
Metric closure	Source incompleteness, anomaly, boundary exchange, or open/effective scope.
Quantum transition closure	Measurement/update, open system, stochastic dynamics, or standing failure.
Cross-response ledger	Source/record laundering.
Lawful redescription	Coordinate, basis, branch, or representative privilege.
No hidden branch selector	Outcome import or many-worlds label laundering.
No hidden horizon selector	Information sink or global-retention laundering.
No undeclared regulator	Cutoff, measure, or renormalization becomes hidden authority.
Reportable scope	Approximation reported as unification theorem.

Part XI

Boundary, Measurement, Entanglement, and Horizon Joint Problems

43 Measurement is not hidden metric sourcing

Measurement update changes quantum standing, record status, or reportability. It cannot be used as a same-scope metric source change without a declared joint ledger.

Theorem 43.1 (No hidden measurement sourcing). *A measurement or collapse event can affect the metric source ledger only as declared update, record, open/effective exchange, or richer-scope structure. It cannot repair closed metric source continuation as the same act.*

44 Decoherence and records

Decoherence is an environment/record ledger. It is not by itself outcome standing, branch reference, or metric source closure.

Proposition 44.1 (Decoherence ledger). *A decoherence account is admissible in a joint GR/QM setting only when the environment, record basis, witness algebra, reduced-state status, and metric source consequences are declared. Otherwise decoherence functions as hidden record laundering.*

45 Entanglement is compatibility, not transport

Entanglement can fix boundary-level compatibility between jointly prepared systems. It does not license hidden metric communication or source repair.

Theorem 45.1 (No entanglement transport laundering). *Entanglement may provide tensor-level compatibility, but it cannot transmit standing-bearing metric source content or repair quantum transition failure without a declared interaction, measurement, record, or boundary ledger.*

46 Horizon is not hidden quantum record sink

Horizon and black-hole information problems are joint-ledger problems. A horizon threatens exterior standing if it functions as a sink for quantum records, source data, or recoverability.

Theorem 46.1 (No horizon record sink). *A horizon cannot serve as a same-scope quantum record sink in a paired GR/QM theorem. If it threatens record or source standing, it must supply exterior witness algebra, leakage/radiation channel, boundary-to-boundary certificate, non-sink quotient, or declared richer-scope structure.*

Proof. A permanent inaccessible sink prevents standing-bearing content from being reused, reported, or lawfully continued in the exterior domain. If the content is physically irrelevant, it is skin. If it is standing-bearing, the model must provide a certificate, leakage channel, non-sink quotient, or richer-scope explanation. Otherwise it is hidden repair or standing failure. $\square$

47 Joint route-closure table

Route	AASC joint status
Measurement collapse changes gravity.	Update/source ledger required.
Decoherence selects source.	Environment/record ledger required.
Entanglement communicates metric influence.	Hidden transport unless interaction/source ledger declared.
Horizon stores information.	Standing sink unless certificate/leakage/non-sink quotient exists.
Renormalized vacuum sources metric.	Vacuum/source ledger and homogeneous quotient discipline required.
Branch weights determine geometry.	Branch/reference/reportability ledger required.
Path-integral measure selects geometry.	Measure/regulator/boundary anchor required.

Part XII

Main Theorems: Structural Unification and Cross-Projection Closure

48 Structural Unification Theorem

Theorem 48.1 (Structural unification of gravity and first-response quantum dynamics). *Let $\mathcal{I}$ be the singleton admissible physical interior fixed by a determinate nondegenerate physical-interior target bearing both metric deformation-response standing and quantum transition/record standing. Then:*

1. $\mathcal{I}$ is unique up to lawful equivalence;
2. target instantiation forces K and A^+ ;
3. gravity is forced as the structural constraint role of $\mathcal{I}$;
4. quantum standing is forced as projective/record-bearing continuation of $\mathcal{I}$;
5. the faithful metric projection yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}};$$

6. the faithful projective-Hilbert projection yields

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi;$$

7. *there is no same-domain split into $\mathcal{I}_g$ and $\mathcal{I}_H$;*
8. *neither projection is primitive over the other;*
9. *cross-projection interaction requires a declared joint ledger.*

Therefore gravity and quantum dynamics at the first-response projective-Hilbert level are structurally unified at the AASC physical-interior level.

Proof. The kernel and A^+ follow from target adequacy. The AASC gravity theorem fixes gravity as the constraint role of admissible physical construction; the Einstein theorem gives its metric first-response projection. The quantum-interior and Schrödinger theorem-use package fixes projective-Hilbert and record-bearing continuation; the Schrödinger theorem gives its scalar nonrelativistic first-response projection. Same-interior fidelity and no plural-interior split make the two projections downstream of the same singleton interior rather than independent applications. Projection non-primacy and backflow exclusion prevent either projection from authorizing the other. When both projections interact, standing changes across the pair must be declared in $\mathcal{L}_{gH}$. Thus the unity is upstream in the physical interior, while the cross-projection problem is downstream ledger closure. $\square$

Remark 48.2 (Scope of structural unification). *The theorem reports structural unification, not completed richer-scope dynamics. It derives the common AASC source of the two projections and the ledger condition for their interaction. It does not compute all quantum-gravity amplitudes, all QFT dynamics, all constants, or all black-hole endpoint behavior.*

49 Cross-Projection Coupling Exhaustion Theorem

Throughout this section, C has the candidate type fixed in Definition 37.1: it may be a same-target rule, term, equation, transition, source-extraction operation, record/update claim, compatibility assertion, boundary/certificate claim, or reconstruction route. The exhaustion theorem therefore applies to the whole class of same-scope cross-projection moves, not only to literal interaction terms in an equation.

Lemma 49.1 (Standing-impact lemma). *Let C be a non-skin cross-projection candidate. Then C changes at least one standing-bearing coordinate: metric source standing, quantum transition standing, record standing, compatibility standing, certificate standing, reportability, or carrier/scope. If it changes none of these, C is skin.*

Proof. A cross-projection candidate has theorem-bearing force only by changing what may be reused or reported in the paired target. The paired target has no fourth same-domain role besides anchors, tensor/operator/record content, and skin. Therefore a non-skin candidate must alter at least one standing-bearing coordinate. If it alters none, no admissible downstream use distinguishes it; it is skin. $\square$

Lemma 49.2 (Metric-impact lemma). *If C changes metric source standing, then the change is admissible only through $\mathcal{L}_g$ as closed metric source content or declared exchange. Without $\mathcal{L}_g$, the candidate is source laundering or metric standing failure.*

Proof. The Einstein-side projection requires a closed or declared-open source ledger. A metric source change that lacks closure, exchange, boundary, anomaly, or effective status inserts content into $T_{\mu\nu}$ without source standing. That is source laundering. If it cannot be licensed even as laundering, the metric side loses standing. $\square$

Lemma 49.3 (Quantum-impact lemma). *If C changes quantum transition standing, then the change is admissible only through $\mathcal{L}_H$ as closed transition standing or declared update/open/effective status. Without $\mathcal{L}_H$, the candidate is hidden update, open/effective scope without declaration, or quantum standing failure.*

Proof. The projective-Hilbert projection distinguishes closed autonomous continuation from update, open, measurement, driven, and effective regimes. A quantum-standing change not typed in $\mathcal{L}_H$ either repairs closed standing by hidden update, omits the open/effective ledger that would license it, or fails the quantum continuation target. $\square$

Lemma 49.4 (Record-impact lemma). *If C uses measurement, collapse, branch selection, decoherence, observer-relative update, or a record as source-relevant or reportable content, then the use is admissible only through*

$$\mathcal{L}_{\text{rec}} = (\text{Rec}, A_{\text{Ref}}/\approx_{\text{MR}}, \text{FixComm}, \text{Trace}, \text{Reuse}, \text{Cert}_{\text{rec}}, \text{Report}).$$

Without $\mathcal{L}_{\text{rec}}$, the candidate is record laundering.

Proof. The measurement-record theorem forces record-fixation assignments and the measurement-record quotient for downstream reuse. It also makes fixation commutation and boundary-trace role fixation the admissible route to record-role hardening once internal and provenance/indexical asymmetries fail. If a candidate uses a record-bearing event but omits these structures, it treats assignment-relative or unfixed record content as if it were source-facing or reportable. That is record laundering. $\square$

Lemma 49.5 (Compatibility-impact lemma). *If C uses entanglement, Bell correlations, boundary compatibility, or nonlocal compatibility as standing-bearing cross-projection content, then the use is admissible only through $\mathcal{L}_{\text{comp}}$ as compatibility. If it is used as metric transport, source transfer, branch selection, communication, or repair, the candidate is compatibility laundering unless retyped as declared cross-ledger or richer scope.*

Proof. The entanglement theorem types Bell correlations as joint standing-class tensor compatibility and closes hidden-variable, signaling, mechanism, and repair typings on the same Bell scope. Thus compatibility may constrain joint standing, but it cannot do transport work. A candidate that makes compatibility move or repair standing has changed role; if undeclared, it launders compatibility into transport. $\square$

Lemma 49.6 (Certificate-impact lemma). *If C relies on horizon, boundary, apparatus, renormalization, regulator, exterior-reconstruction, or endpoint claims, then the relevant certificate must be declared in $\mathcal{C}_{\text{cert}}$. Without certification, the candidate is certificate laundering or standing failure.*

Proof. Those claims alter what may be reused at the boundary of a projection or at a threatened endpoint. Such alteration is standing-bearing. If the certificate is declared, the claim is ledger content; if it is not declared, the claim uses unlicensed boundary authority or repairs standing after failure. $\square$

Lemma 49.7 (Scope-impact lemma). *If C changes carrier, dimension, topology, field content, operator algebra, quantization grammar, dynamics, measurement regime, or locality envelope, then C is declared richer/open/effective scope rather than same-scope coupling. If the change is undeclared, it is hidden scope transfer.*

Proof. Carrier and envelope changes alter the theorem object. They may be legitimate in declared richer scope, but they cannot function as undeclared same-scope alternatives. Treating them as same-scope is scope transport. $\square$

Lemma 49.8 (Skin/failure dichotomy). *After metric, quantum, record, compatibility, certificate, and scope impacts are classified, any remaining cross-projection candidate either changes no standing-bearing coordinate and is skin, or claims standing without a ledger and fails.*

Proof. The standing-impact lemma exhausts the coordinates on which a non-skin candidate can act. The preceding lemmas classify each coordinate. A candidate left outside those classes either changes nothing standing-bearing or asserts force without admissible support. $\square$

Theorem 49.9 (Cross-projection coupling exhaustion). *For every attempted same-scope coupling C between the Einstein and Schrödinger projection normal forms,*

$$\text{SameScopeJoint}_1(C) \Rightarrow C \in \mathcal{L}_{gH}.$$

If $\text{SameScopeJoint}_1(C)$ fails, then C falls into at least one of the following typed failure/scope routes: projection backflow, source laundering, hidden update or quantum standing failure, record laundering, compatibility laundering, certificate laundering, open/effective scope, measurement/update scope, richer-scope joint continuation, skin, or standing failure. No unclassified same-scope coupling remains. Multiple failure routes may co-occur; a candidate is assigned a primary failure type by the first standing-bearing role it attempts to perform without declaration, and co-occurrence strengthens rather than weakens the exhaustion result.

Proof. If $\text{SameScopeJoint}_1(C)$ holds, every standing-bearing role of C is carried by a declared component of $\mathcal{L}_{gH}$, so C is a joint-ledger term. If the predicate fails, the standing-impact lemma applies unless C is skin. Metric impact is classified by the metric-impact lemma; quantum impact by the quantum-impact lemma; record impact by the record-impact lemma; compatibility impact by the compatibility-impact lemma; certificate impact by the certificate-impact lemma; and carrier/envelope changes by the scope-impact lemma. The skin/failure dichotomy closes the residual cases. These cases are mutually typed by the role the candidate attempts to perform, and together they exhaust same-scope cross-projection coupling attempts. $\square$

50 AASC-free reconstruction boundary

A non-AASC reconstruction route is relevant to this manuscript only if it proves the same paired target with fewer declared anchors and without hidden selector, source, record, boundary, or continuation import, or if it identifies a failed named AASC link. Otherwise it is a parallel program, not a refutation.

Proposition 50.1 (Separate reconstruction route). *Canonical, path-integral, semiclassical, emergent-spacetime, or other quantum-gravity programs may corroborate, refine, or replace parts of the paired-projection theorem only by addressing the same target and preserving the same standing, reference, admissibility, irreversibility, redescription, continuation, and reportability burdens.*

51 Correctness-relevant denial targets

A same-mode denial is correctness-relevant only if it identifies one of the following failures:

1. target adequacy does not force the kernel;
2. singleton admissible-interior closure fails;
3. the no-plural-interior split theorem fails;
4. the physical interior does not require quotient architecture;
5. the Einstein theorem-use package fails in its declared scope;
6. the Schrödinger theorem-use package fails in its declared scope;
7. paired projections can authorize each other without hidden backflow;
8. cross-projection influence can carry standing without a joint ledger;
9. semiclassical expectation values are automatically closed metric sources;
10. measurement/update can repair closed source and transition standing as the same act;
11. horizons can serve as standing sinks without certificate;
12. modified GR/QM interface routes have been denied rather than typed;
13. the theorem overreports beyond paired first-response projection and joint-ledger necessity.

Part XIII

Conclusion

AASC does not reconcile GR and QM by reducing one to the other. It shows why both arise from the same singleton admissible physical-interior structure. Gravity is structurally necessary, not a quantum target. Quantum dynamics is a faithful projection, not the ground of gravity. The unity is not imposed by a joint formalism after the fact; it is forced by singleton admissible-interior closure.

The proof has three layers. First, target instantiation forces the kernel and the fixed-domain consequence layer. Second, the admissible physical interior forces two projection roles: gravity as structural constraint necessity and quantum standing as projective/record-bearing continuation. Third, the faithful first-response projections yield the two familiar normal forms:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}},$$

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m}\Delta + M_V\right)\psi.$$

They are paired theorem objects of one admissible interior, not primitive rivals and not outputs of two externally coordinated interiors.

The joint-ledger theorem then states the downstream interaction discipline. Cross-projection use must preserve metric source closure, quantum transition standing, admissible record fixation, boundary compatibility, lawful redescription, joint continuation, and reportability through a declared ledger. Without that ledger, a purported coupling is projection backflow, source laundering, record laundering, compatibility laundering, measurement/update scope, open/effective scope, boundary/certificate route, richer scope, or standing failure.

Quantum gravity, properly typed, is the problem of richer-scope joint continuation preserving both projections. It is not the primitive quantization of gravity, because gravity is already the structural constraint role of admissible physical construction, and it is not generation of a second admissible interior, because same-domain plural interiors are closed. Canonical, path-integral, semiclassical, emergent-spacetime, holographic, collapse, and QFT-on-curved-spacetime programs are therefore not rejected; they are audited by the ledger they actually carry.

The final report is:

The unity is upstream; the cross-projection problem is downstream.
--

Equivalently: gravity and first-response projective-Hilbert quantum dynamics are not externally reconciled; they are paired projections of the one admissible interior, with richer quantum regimes typed as declared scope enrichments.

Appendices

A Source Theorem-Use Ledger

This appendix consolidates theorem-use provenance for the structural-unification manuscript. It records how each source enters the proof without reproducing its full internal development. The main text remains the proof carrier for the present theorem; the source papers provide stable theorem-use modules and audit expansion.

Source cluster	DOI	Theorem-use role
Kernel and admissibility	10.5281/zenodo.19324199	Target adequacy, kernel necessity, no optional framework, and no same-act repair.
Singleton admissible interior	10.5281/zenodo.18518484	Uniqueness of the admissible interior, elimination of discrete plural interiors, no competing gates, no carriers/generators/repairs, and no gravity/quantum interior split.
Gravity as structural necessity	10.5281/zenodo.19945401	Gravity as the admissibility-forced constraint role of the physical interior, prior to metric dynamics and prior to quantization.
Einstein dynamics	10.5281/zenodo.20538579	Faithful metric projection, SameScopeMetric ₁ ($E_{\mu\nu}$), curvature-response exhaustion, closed source continuation, and Einstein normal form.
Schrödinger dynamics	10.5281/zenodo.21254589	Faithful projective-Hilbert projection, Wigner-Stone continuation, SameScopeQuantumSpatial ₁ (H), Hamiltonian-response exhaustion, and Schrödinger normal form.
Measurement records	10.5281/zenodo.18514647	Record-fixation quotient, fixation commutation, boundary-trace record certification, selector exhaustion, and reportable downstream reuse.
Entanglement compatibility	10.5281/zenodo.19401432	Bell witness fixation, joint standing class, tensor compatibility, no hidden variables/signaling/mechanism/repair, and boundary compatibility rather than transport.
Claim standing and legitimacy	10.5281/zenodo.20312075	Reportability, laundering exclusion, target-carrier discipline, and non-overreporting of effective or richer-scope claims.

Anchor, tensor, and skin	10.5281/zenodo.18678041	Role typing of anchors, standing-bearing tensor/operator/compatibility content, and presentation skin.
Operator closure	10.5281/zenodo.19183407	Same-scope operator and auxiliary-datum closure discipline for the Schrödinger and joint-coupling predicates.
Self-adjoint extension selection	10.5281/zenodo.20678826	Domain and boundary-condition certification for Hamiltonian and source-extraction ledgers.
Realized physical interior	10.5281/zenodo.20129729	Standard-physics-stack bridge and projection discipline for realized physical structures.

Theorem-use discipline. The source ledgers are not optional interpretive supplements. They are theorem-use modules in the declared AASC proof class. The present manuscript uses them in restricted ways: singleton closure blocks plural interiors; the Einstein and Schrödinger papers supply the paired first-response normal forms; the measurement paper supplies record certification; the entanglement paper supplies compatibility-not-transport closure; and UEAP/ATS sources control reportability and role typing. Claims beyond those theorem-use roles require richer-scope declaration.

B Formal Predicates and Cross-Projection Exhaustion

B.1 Cross-projection candidate type

A cross-projection candidate is any proposed same-target rule, term, equation, transition, source-extraction operation, record/update claim, compatibility claim, boundary/certificate claim, or reconstruction route that changes or purports to change standing-bearing content across the metric and projective-Hilbert projections. For audit one may write

$$C = (C_g, C_H, C_{\text{cross}}, C_{\text{rec}}, C_{\text{comp}}, C_{\text{cert}}, C_{\text{cont}}, C_{\text{rep}}),$$

where each coordinate is empty, inert, declared, or standing-bearing. The coordinates record metric impact, quantum impact, cross-response impact, record/update impact, compatibility impact, certificate impact, continuation impact, and reportability impact.

B.2 The predicate $\text{SameScopeJoint}_1(C)$

Write $\text{SameScopeJoint}_1(C)$ iff all of the following hold:

1. C is internal to the same singleton physical-interior target and does not split $\mathcal{I}$ into $\mathcal{I}_g$ and $\mathcal{I}_H$.
2. C preserves projection non-primacy: neither projection authorizes, generates, selects, or repairs the other.
3. Any metric-standing effect of C is carried by $\mathcal{L}_g$ as closed source content or declared exchange.
4. Any quantum-standing effect of C is carried by $\mathcal{L}_H$ as closed transition standing or declared update/open status.

5. Any measurement, branch, record, collapse, decoherence, or outcome content used by C is certified by $\mathcal{L}_{\text{rec}}$.
6. Any entanglement, Bell, boundary, or nonlocal compatibility content used by C is certified by $\mathcal{L}_{\text{comp}}$ and is not retyped as transport.
7. Boundary, horizon, renormalization, apparatus, exterior-reconstruction, and regulator roles are declared in $\mathcal{C}_{\text{cert}}$ when they carry standing.
8. C is lawful under metric redescription and projective-Hilbert/ray redescription.
9. C imports no hidden branch selector, source selector, record selector, compatibility transport, or scope transfer.
10. C reports only the scope carried by Rep.

B.3 Cost of weakening SameScopeJoint₁

Clause weakened	Cost
Same singleton interior	The coupling becomes external coordination, plural-interior split, or scope change.
No projection backflow	A downstream projection authorizes upstream standing or another projection.
Metric ledger	Source laundering, nonconservation, or metric source-standing failure.
Quantum ledger	Hidden update, open/effective dynamics, or quantum standing failure.
Record ledger	Measurement, branch, outcome, collapse, or decoherence content is record-laundered.
Compatibility ledger	Entanglement is misread as transport, source transfer, signaling, or repair.
Certificate ledger	Horizon, boundary, renormalization, apparatus, or exterior claims become uncertified selectors.
Lawful redescription	Coordinates, bases, branch labels, or presentation skin acquire standing-bearing force.
No hidden selectors	Branch, source, record, horizon, or regulator choice is imported without standing.
Reportability	Effective or mixed-scope material is overreported as structural unification.

B.4 Exhaustion lemma chain

Lemma B.1 (Standing-impact lemma). *Every non-skin cross-projection candidate changes at least one standing-bearing coordinate: metric standing, quantum standing, record standing, compatibility standing, certificate standing, continuation status, reportability, or carrier/scope.*

Proof. If none of those coordinates changes, the candidate has no admissibility-bearing effect on the paired theorem object. It is therefore skin. If any coordinate changes, it falls under the corresponding ledger coordinate by definition of the paired target. $\square$

Lemma B.2 (Metric-impact lemma). *Any metric-standing effect without $\mathcal{L}_g$ is source laundering, declared open/effective scope, or metric standing failure.*

Proof. Metric standing is source-facing in the Einstein projection. A metric effect must either be closed, have declared exchange, be explicitly effective/open, or lack standing. Without $\mathcal{L}_g$, no admissible source ledger carries the effect. $\square$

Lemma B.3 (Quantum-impact lemma). *Any quantum-standing effect without $\mathcal{L}_H$ is hidden update, open/effective scope, or quantum standing failure.*

Proof. The quantum projection requires closed transition standing or declared update/open status. An effect that changes quantum standing without that ledger either imports update work secretly, declares an open/effective target, or fails to preserve the quantum theorem object. $\square$

Lemma B.4 (Record-impact lemma). *Any measurement, update, branch, record, outcome, collapse, or decoherence use without $\mathcal{L}_{\text{rec}}$ is record laundering.*

Proof. Record-bearing use requires the measurement-record quotient, fixation commutation, boundary trace or declared selector locus, downstream reuse, certification, and reportability. Without those, the event is used as if it had a fixed record role while no such role has standing. $\square$

Lemma B.5 (Compatibility-impact lemma). *Any use of entanglement, Bell correlation, boundary compatibility, or nonlocality as transport, source transfer, horizon rescue, record selector, or same-scope repair without $\mathcal{L}_{\text{comp}}$ is compatibility laundering.*

Proof. The Bell-scope theorem-use package types entanglement as joint standing-class tensor compatibility. It does not authorize metric transport, hidden variable completion, signaling, mechanism, or repair. Any such use changes the role unless declared in a richer cross-ledger. $\square$

Lemma B.6 (Certificate-impact lemma). *Horizon, boundary, renormalization, regulator, apparatus, exterior-reconstruction, and domain claims without $\mathcal{C}_{\text{cert}}$ are certificate laundering or scope change.*

Proof. Each listed claim changes which content may be reused or sourced across the paired target. If no certificate fixes that role, the claim imports selector work. If the role is declared, the candidate is richer, effective, boundary, or certified scope rather than an undeclared same-scope coupling. $\square$

Lemma B.7 (Scope-impact lemma). *Changes of carrier, topology, field content, dimension, connection type, quantization grammar, measurement/update status, or dynamics are richer/open/effective scope unless already declared inside $\mathcal{L}_{gH}$.*

Proof. Such changes alter anchors or tensor/operator load rather than preserving the same target. They are legitimate when declared, but they are not hidden same-scope alternatives. $\square$

Theorem B.8 (Cross-projection coupling exhaustion). *If SameScopeJoint₁(C) holds, then C is a declared joint-ledger term, $C \in \mathcal{L}_{gH}$. If SameScopeJoint₁(C) fails, then C falls into at least one typed failure/scope route: projection backflow, source laundering, hidden update, record laundering, compatibility laundering, certificate laundering, measurement/update scope, open/effective scope, richer-scope joint continuation, skin, or standing failure. No unclassified same-scope coupling remains.*

Proof. By the standing-impact lemma, a non-skin candidate must change at least one standing-bearing coordinate. Metric effects are classified by the metric-impact lemma; quantum effects by the quantum-impact lemma; record/update effects by the record-impact lemma; compatibility effects by the compatibility-impact lemma; boundary, horizon, renormalization, apparatus, and reconstruction effects by the certificate-impact lemma; and carrier or dynamics changes by the scope-impact lemma. If all standing-bearing coordinates are declared and compatible, C is precisely a joint-ledger term. If at least one coordinate is undeclared or incompatible, the corresponding typed failure/scope route applies. Multiple routes may co-occur; the primary type is the first standing-bearing role the candidate attempts to perform without declaration. Co-occurrence strengthens rather than weakens the exhaustion. $\square$

C Worked GR/QM Interface Audits

Each audit uses the same template: target expression, required ledgers, pass conditions, failure routes, and reportable status.

C.1 Semiclassical gravity

Target expression:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}}.$$

Required ledgers: a metric ledger $\mathcal{L}_g$, a quantum state/algebra ledger $\mathcal{L}_H$, an expectation-value extraction ledger $\mathcal{L}_{\text{cross}}$, a renormalization certificate, record/update status when ω depends on measurement or branch data, and divergence closure or declared exchange.

Theorem C.1 (Semiclassical pass/fail audit). *Semiclassical gravity is admissible as an effective mixed-projection ledger iff the state ω , algebra/domain, renormalization prescription, record/update status, and divergence closure are certified. If any of those roles is standing-bearing but undeclared, the model is source laundering, record laundering, open/effective scope without declaration, or standing failure.*

Proof. The Einstein side requires a closed or exchange-declared source. The expectation value $\langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}}$ supplies such a source only after state, algebra/domain, and renormalization are fixed. If ω depends on measurement, branch, collapse, or record content, $\mathcal{L}_{\text{rec}}$ is required. If $\nabla^{\mu} \langle \hat{T}_{\mu\nu} \rangle_{\omega}^{\text{ren}} = 0$ fails without declared exchange, the metric ledger is not closed. When all certificates are supplied, the model is a reportable effective mixed-projection ledger; without them, it launders source or record standing. $\square$

Reportable status: effective mixed-projection theory when certified; not the full structural unification itself.

C.2 QFT on curved spacetime

Target expression: quantum fields over a fixed Lorentzian metric background. Required ledgers: fixed metric anchor, field algebra, state choice, Hadamard/renormalization conditions where relevant, and reportable background scope. Pass condition: the metric is treated as an anchor and backreaction is not overreported. Failure route: if backreaction is claimed, a joint ledger is required; otherwise metric source effects are laundered.

C.3 Canonical, Wheeler-DeWitt, and constraint quantization routes

Target expression: projective-Hilbert or related representation of metric/constraint degrees of freedom. Required ledgers: constraint algebra, quotient/gauge structure, time/continuation rule, observables, record status, and source/metric closure. Pass condition: declared richer-scope joint continuation that preserves gravity as structural constraint role. Failure route: Hilbert backflow into the gravity role or unresolved continuation standing.

C.4 Path-integral and histories routes

Target expression: integral or sum over metric, matter, or history presentations. Required ledgers: target, measure, gauge fixing, regulator, boundary data, record status, and reportable scope. Pass condition: all standing-bearing choices are declared. Failure route: hidden measure, regulator, or boundary selector.

C.5 Collapse, stochastic gravity, and metric source change

Target expression: update dynamics plus metric-facing source consequences. Required ledgers: update/dynamic-selection status, record-fixation ledger, metric source ledger, stochastic/exchange ledger when applicable. Pass condition: source closure or declared exchange after record certification. Failure route: hidden update, source laundering, or same-act repair.

C.6 Holography, emergent spacetime, horizons, and nonlocal routes

Target expression: metric reconstruction or information preservation from boundary, Hilbert, entanglement, or exterior data. Required ledgers: reconstruction map, boundary/exterior certificate, compatibility ledger, non-sink quotient or leakage/radiation channel, and reportability. Pass condition: declared richer-scope reconstruction or certificate route. Failure route: projection backflow, compatibility laundering, horizon standing sink, or uncertified boundary claim.

D Record and Compatibility Route Closure

D.1 Measurement-record route closure

The measurement-record theorem-use ladder is:

$$\text{record-bearing object} \Rightarrow A_{\text{Ref}}/\approx_{MR} \Rightarrow \text{FixComm} \Rightarrow \text{Trace} \Rightarrow \text{Reuse} \Rightarrow \text{Report}.$$

The corresponding joint-ledger coordinate is

$$\mathcal{L}_{\text{rec}} = (\text{Rec}, A_{\text{Ref}}/\approx_{MR}, \text{FixComm}, \text{Trace}, \text{Reuse}, \text{Cert}_{\text{rec}}, \text{Report}).$$

No metric-facing measurement source is admissible without record-fixation quotient and fixation commutation. If a collapse, branch, outcome, decoherence, detector, or apparatus event is used as source content without $\mathcal{L}_{\text{rec}}$, the candidate performs record laundering or changes scope.

D.2 Entanglement compatibility route closure

The entanglement theorem-use ladder is:

$$\begin{aligned} \text{Bell witness fixation} &\Rightarrow \text{joint standing class} \Rightarrow \text{tensor compatibility} \\ &\Rightarrow \text{no hidden variable/signaling/mechanism/repair} \Rightarrow \text{boundary compatibility}. \end{aligned}$$

The corresponding joint-ledger coordinate is

$$\mathcal{L}_{\text{comp}} = (\text{Wit}_{\mathcal{B}}, \text{JointStand}, \text{TenComp}, \text{Trace}_{\mathcal{B}}, \text{NoTransport}).$$

Boundary compatibility is not interior metric transport. It cannot by itself serve as source transfer, horizon information rescue, branch selector, measurement record selector, or same-scope repair. Such uses require a declared cross-ledger, certificate route, or richer scope.

E Denial Costs and Adversarial Audit Matrix

E.1 Kernel, singleton, and projection denial costs

Denied item	Cost
Kernel necessity	No determinate paired theorem object.
Singleton admissible interior	The manuscript collapses into external coordination of two regimes rather than structural unification.
No plural-interior split	Gravity and quantum standing are assigned to distinct domains; same-domain unification is lost.
Projection non-primacy	One projection becomes a primitive selector or ground for the other.
No quantization primacy	The projective-Hilbert projection is used to generate the gravity role that licensed the metric projection.
No projection backflow	Downstream expressions authorize upstream standing.

E.2 Joint-ledger denial costs

Ledger denied	Cost
$\mathcal{L}_g$	No metric source closure; Einstein-side source standing fails.
$\mathcal{L}_H$	No quantum transition-standing or state/update identity.
$\mathcal{L}_{\text{cross}}$	No coupling; only parallel projections or hidden backflow remain.
$\mathcal{L}_{\text{rec}}$	Measurement/update cannot become reportable source or evidence.
$\mathcal{L}_{\text{comp}}$	Entanglement is misread as transport, hidden mechanism, or repair.
$\mathcal{C}_{\text{cert}}$	Record, source, boundary, horizon, or renormalization claims become uncertified.
$\text{Cont}_{\text{joint}}$	No unified continuation; only mixed-scope fragments remain.
Rep	Approximation or effective model is overreported as unification.

E.3 Adversarial audit matrix

Objection	Standing?	Response
AASC is optional vocabulary.	No, unless a kernel link fails.	Determinate paired target already requires admissibility, standing, reference, and irreversibility.
This is not unification.	No.	Singleton interior closure plus paired projections supply structural unification; joint ledger handles cross-projection interaction.
The paper coordinates two independent interiors.	No, unless singleton closure fails.	No plural-interior split remains on one fixed domain.
Gravity should be quantized.	No as stated.	Gravity is structural constraint necessity; quantization may be a richer-scope joint continuation route, not primitive authorization.
The equations are assumed.	No if theorem-use packages stand.	Einstein and Schrödinger papers provide the paired normal-form proofs.
Semiclassical gravity is enough.	Only if certified.	It is an effective mixed-projection ledger iff state, renormalization, record/update status, and divergence closure are certified.
Entanglement gives hidden nonlocal transport.	No.	Bell-scope compatibility is tensor compatibility, not metric transport or signaling.
Measurement collapse supplies metric source.	Only if record/source certified.	Otherwise it is record/source laundering or declared update scope.
The failure taxonomy overlaps.	Not an objection.	Candidates may fail in multiple coordinates; the theorem requires at least one typed route, not exactly one.
The result solves all quantum gravity.	No.	It reports structural unification and cross-ledger closure, not an exhaustive richer-scope dynamics.

F Journal Compression Map

This project version keeps the full audit scaffolding. A journal-length version can preserve theorem force while compressing supporting material as follows.

Full manuscript material	Journal disposition
Proof spine and structural-unification theorem	Keep in main text.
Kernel and singleton interior closure	Keep compressed in main text.

Einstein and Schrödinger theorem-use ladders	Keep as theorem-use summary with DOI citations.
SameScopeJoint ₁ predicate and exhaustion lemma chain	Keep in main text.
Worked interface audits	Keep semiclassical audit in main text; move others to supplement.
Record and compatibility route closure	Keep compressed; move detailed ladder to supplement.
Denial-cost and adversarial matrices	Move to supplement unless journal permits audit appendices.
Source theorem-use ledger	Keep shortened provenance table.

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